

**A Multimodal Deepfake Detection Framework Leveraging Bilinear Pooling and Emotion
Mismatch**

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Chapter 1

THE PROBLEM AND ITS SETTING

Introduction

The rapid development and adoption of generative AI technologies have revolutionized digital media and forced us into a world where it becomes increasingly harder to distinguish between real and fake media. However, at the forefront of this revolution are Deepfakes, which are realistic media produced using deep learning that allows for face and voice swaps with unprecedented ease (Adinarayana et al., 2026). While they hold tremendous potential for entertainment and education, they also usher in a new age of digital deception. The problem lies not in the technology itself, but in how easily it has become available and the ways in which it is used. What once required government-level tools or expert knowledge can now be done by anyone with a laptop and wi-fi, letting people create fake but convincing videos, images, and audio. Deepfakes pose a uniquely dangerous threat to information integrity, reputation, and cybersecurity. Unlike traditional media manipulation, deepfakes weaponize the human face and voice, the two signals people instinctively trust above all others (Mittal et al., 2020). This is why they are particularly effective as a medium for political misinformation, non-consensual generative images, and biometric spoofing—a process of tricking a biometric authentication system using a fake biometric sample (Banywana, 2026).

The biggest problem in detecting deepfakes currently is how far they have come in development. Earlier versions were impaired by glitches and errors that made them easily detectable, but current versions are almost indistinguishable from real media (Gambin et al., 2024). In response, early detection systems adopted a unimodal approach, analyzing either the visual and audio stream in isolation. While these systems proved effective against earlier deepfakes, their accuracy is tied directly to the generation method they were trained on, and

they fail to capture the cross-modal coherence that exists between the different channels of genuine human communication (Cai et al., 2023; Heidari et al., 2023). This is what multimodal deepfake detection frameworks aim to solve, rather than analyzing a single stream, multimodal approaches combine visual, audio, and textual signals to check whether the emotional and semantic content across all channels is consistent with one another (Javed et al., 2025). However, most existing multimodal detection frameworks still rely heavily on visual artifacts, such as blending seams, unnatural skin textures, and resolution inconsistencies left behind by the generation process, as their primary detection basis. But as generative models evolve, particularly with the rise of diffusion-based synthesis, a newer generation of AI that learns to generate highly realistic media by studying patterns from massive amounts of real images and videos, resulting in cleaner outputs with fewer detectable artifacts, these detectors experience accuracy degradation (Rossler et al., 2019; Corvi et al., 2023). Furthermore, the benchmark datasets they are trained on are largely lab-generated, limiting their ability to generalize against real-world deepfakes due to the lack of in-the-wild training (Dolhansky et al., 2020; Zi et al., 2020).

In order to address these challenges, we apply bilinear pooling in order to incorporate the visual, audio, and text features in our approach and thus capture their interaction for detecting emotional discrepancies. We train our network on emotional inconsistency examples extracted using data augmentation techniques applied to the CREMA-D and MELD datasets, with authentic real-world samples sourced from CMU-MOSEI. Testing is done using FakeAVCeleb, an audio-video multimodal deepfake detection benchmark featuring both face-swap and voice-conversion manipulations.

Theoretical Framework

This study is grounded in three interconnected bodies of theory that collectively justify the design and approach of the proposed detection framework: Multimodal Emotion Recognition Models and Representation Learning, Deepfake Detection Frameworks, and Multimodal Fusion via Bilinear Modeling.

1. Multimodal Emotion Recognition Models and Representation Learning

Multimodal Emotion Recognition (MER) models provide the technical foundation for how emotional signals are extracted and interpreted across distinct modalities in this study. Unlike unimodal models that analyze a single channel, MER models combine information from multiple sources – such as facial expressions, vocal tone, and spoken language – to build a more complete and accurate representation of a person’s emotional state (Abdullah et al., 2021). This is directly relevant to the research problem because the ability to detect deepfakes through emotional inconsistency depends first on having reliable models that can independently recognize and classify emotion within each modality before cross-modal comparisons can be made. Additionally, current generative AI models, such as GANs and diffusion models, are designed to produce convincing fakes within their respective modalities but may not be optimized to coordinate emotion across all of them. (Hosler et al., 2021). As a result, a deepfake face may carry a neutral expression while the voice expresses fear, or the spoken words may convey joy while the face remains still (Dang et al., 2020)

In this study, three specialized architectures serve as the backbone of multimodal emotion recognition: Wav2Vec 2.0 for vocal emotion, BERT for linguistic sentiment, and Vision Transformer (ViT) for facial expression. What makes these architectures particularly well-suited for this task is their reliance on representation learning, the ability to automatically learn rich, meaningful features directly from raw data without requiring manual feature engineering. Rather than depending on hand-crafted inputs such as spectrograms or predefined facial landmarks,

each model learns its own deep representation of emotionally relevant patterns from large-scale training data. Wav2Vec 2.0 applies self-supervised representation learning to extract prosodic and emotional features directly from raw audio waveforms, capturing subtle cues such as pitch, energy, and speaking rate that reflect a speaker's emotional state (Hsu et al., 2021). BERT uses bidirectional representation learning to understand not just individual words but their full semantic and sentiment context, capturing the emotional meaning of what is being said rather than just the words themselves. ViT learns patch-level spatial representations of facial images through its self-attention mechanism, capturing both local muscle configurations and global facial relationships that together convey emotional state (Chen et al., 2022). Wav2Vec 2.0 and BERT are jointly supervised through a shared emotion classification head on their combined audio-text representation, while the ViT is supervised by a separate visual emotion head, before their outputs are compared and fused. This multi-task design is grounded in the theoretical principle that genuine human emotion produces consistent and correlated signals across all three channels simultaneously (Mehrabian, 1971).

2. Deepfake Detection Frameworks

Deepfake detection frameworks describe the architectural and methodological approaches that have been developed to identify synthetic media. Early detection frameworks were built around the identification of visual artifacts, low-level anomalies such as blending seams, unnatural skin textures, and resolution inconsistencies that early generative models left behind in their output. Convolutional neural networks (CNN) became a standard tool for this task, as its ability to detect fine-grained spatial patterns made them well-suited for identifying traces of face manipulation (Gong et al., 2021). These artifact-based frameworks, however, carry a fundamental weakness: their performance is tied directly to the generation method they were trained on. As generative models advance, particularly with the rise of diffusion-based synthesis that produces photorealistic output with far fewer detectable artifacts,

artifact-dependent detectors experience significant accuracy degradation (Rossler et al., 2019; Corvi et al., 2023).

In response, a second generation of detection frameworks shifted focus from image quality to behavioral authenticity. Tolosana et al. (2020) demonstrated that temporal inconsistencies in facial movement could serve as a more reliable detection signal, while Mittal et al. (2020) extended this further by identifying audiovisual mismatches based on lip synchronization error. These behavioral frameworks marked a significant step forward because they target signals that are harder for generative models to eliminate. This study builds upon and extends this behavioral detection logic. Rather than relying on lip synchronization alone, this framework centers detection on cross-modal emotional inconsistency — the disagreement between the emotion expressed in the face against the combined signal of the voice and spoken words.

3. Multimodal Fusion via Bilinear Modeling

Multimodal Fusion Theory addresses the question of how information from independent input channels can be combined into a unified representation. Baltrusaitis, Ahuja, and Morency (2019) provide the canonical terms of fusion strategies, early, late, and intermediate, and observe that intermediate fusion tends to be well-suited for tasks where the discriminative signal lies in the relationship between modalities rather than within any single modality. Deepfake detection through emotional inconsistency aligns with this condition, as the signal of cross-modal mismatch resides in the relationship between audio and visual features rather than within either modality in isolation.

Within intermediate fusion, this study employs bilinear pooling, a technique whose theoretical foundations were established by Tenenbaum and Freeman (2000) for separating style and content via bilinear models, later adapted for deep learning by Lin, RoyChowdhury,

and Maji (2015) for fine-grained visual recognition through dual-stream bilinear CNNs, and extended to cross-modal settings by Fukui et al. (2016) for multimodal vision-language tasks. The mathematical principle underlying bilinear pooling is the outer product of two feature vectors, which produces a matrix containing every pairwise multiplicative interaction between elements of the two vectors. Formally, for an audio-text feature vector $a \in \mathbb{R}^m$ and a visual feature vector $v \in \mathbb{R}^n$, the bilinear product $f = a \otimes v \in \mathbb{R}^{m \times n}$ encodes $m \times n$ distinct interaction terms, which can then be flattened into a vector for classification, with each interaction independently weighted by the downstream model.

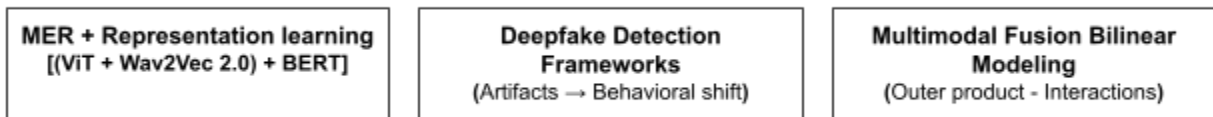
Prior work has shown, both empirically and through architectural analysis, that bilinear pooling can offer advantages over simple concatenation for multimodal tasks where the discriminative signal resides in feature interactions rather than in individual feature values (Lin, RoyChowdhury, & Maji, 2015; Fukui et al., 2016). Concatenation, $[a; v]$, preserves the original feature dimensions but provides no explicit mechanism for the classifier to directly observe interactions between specific audio and visual features. A neural network operating on concatenated features must learn cross-modal interactions implicitly through hidden layer weights, an indirect path that provides no architectural guarantee of capturing the relevant multiplicative structure. The outer product, by contrast, explicitly materializes every possible interaction as a distinct feature dimension, allowing the classifier to learn that a specific combination, for example, an audio dimension corresponding to angry prosody activating alongside a visual dimension corresponding to a happy expression, can be indicative of emotional inconsistency. This is conceptually similar to the operation Tenenbaum and Freeman (2000) used to separate style from content: in their formulation, the bilinear product helps disentangle two factors of variation, and in this study's formulation, it supports the disentanglement of emotional content from modality of expression. By framing emotional mismatch as a learnable function over the bilinear interaction space, this study grounds its

fusion choice in a mathematical principle — why the outer product captures cross-modal interactions — rather than in empirical preference, where a method is selected because it performs well in benchmarks rather than because its structure aligns with the problem.

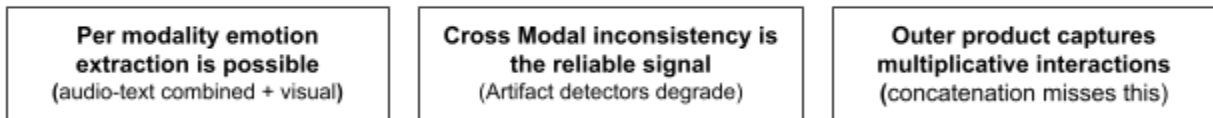
Conceptual Framework

Figure 1. Conceptual Framework

Theoretical bodies



What each theory contributes



Detection framework



The framework is established based on three related theoretical constructs. Firstly, the theoretical construct of Multimodal Emotion Recognition and Representation Learning states that emotions can be correctly recognized using different modalities due to the self-supervised learning methodology created using huge volumes of data. Secondly, according to Deepfake Detection Frameworks, cross-modal behavior inconsistency represents a more effective detection method than visual inconsistencies since the latter loses its relevance with the development of advanced generative methods. Thirdly, the theoretical construct of Multimodal Fusion via Bilinear Modeling claims that intermediate fusion of modalities is carried out through the outer product. MER and representation learning provides a specific functional rationale for

the incorporation of Wav2Vec 2.0, BERT, and ViT as per-modality emotion encoders, where the choice of each neural architecture is explained by its ability to learn self-supervised features, rather than through feature engineering. Behavioral detection theory provides a rationale for focusing the detection of the fake signal on emotional incongruity between two modalities, providing an explanation for why artifact-based methods will not suffice for the generalization criteria in this study. Bilinear fusion method provides a rationale for performing an outer product of the two modalities at the fusion level.

All three theories come together under one theory where Wav2Vec 2.0, BERT, and ViT respectively encode the emotions in each modality, and the bilinear pooling method combines the audio-text combined and visual feature vectors by multiplying the vectors in all possible ways. This theory stems from the idea that true expressions will always produce emotions in all the modalities in a consistent manner. If there is no consistency in the emotional content across all the modalities, then the expression is fake.

Variables in this experiment include the information flow within the process of detecting deepfakes, starting with the video clip input and ending with the classification result output.

- An **independent variable** in this experiment is whether the clip input is genuine human expression or deepfake. This variable is known beforehand during the process of preparing datasets for the experiment.
- A **dependent variable** includes the output value produced by the neural network, which includes a binary classification of deepfake authenticity, with a confidence score between zero and one.

These include the following mediating variables in the detection chain:

- **Modality feature vectors** - The high dimensional vectors derived through Wav2Vec 2.0 (audio), BERT (text) and ViT (visual) from the preprocessed clip

- **Emotion head output vectors** - The six-class probability vectors derived from the emotion head of each modality that indicates the prediction of emotional states by each modality
- **Discrepancy score** – The absolute difference between the emotion head output vectors of audio and visual modality; acts as the key variable for detecting emotional dissonance
- **Bilinear fusion vector** – The outer product of the vectors in the audio and visual modalities to incorporate all interactions prior to classification
- **Keyframe expressiveness scoring formula** — fixed as the product of RetinaFace detection confidence, Laplacian sharpness, and Action Unit saliency ($\text{conf} \times \text{sharpness} \times \text{AU}_{\text{saliency}}$), used to select the eight most expression-relevant frames prior to visual encoding

The following are held at a constant as controlled variables

- **Audio sample rate** — fixed at 16kHz (required by Wav2Vec 2.0 architecture)
- **Temporal sampling rate** — fixed at 25 frames per second before K selection
- **Number of keyframes (K)** — fixed at 8 per clip
- **Visual resolution** — fixed at 224×224 pixels per frame
- **Clip length** — CREMA-D and MELD use source-defined utterance boundaries; CMU-MOSEI clips are filtered to 2 to 8 second intervals
- **Emotion taxonomy** — six classes across all datasets: anger, disgust, fear, happiness, sadness, and neutrality
- **Classification threshold** — fixed at 0.5 for binary real/fake output

Statement of the Problem

With the advancement of artificial intelligence, deepfake generation has reached a level of sophistication where synthetic media is nearly indistinguishable from authentic content. Existing detection frameworks rely primarily on visual artifacts as their detection basis, making them vulnerable to accuracy degradation as generative models continue to evolve. Furthermore, most detection systems analyze only a single modality in isolation, failing to exploit the cross-modal emotional inconsistencies that current deepfake generation pipelines have yet to fully address.

Research Questions

This study proposes a multimodal deepfake detection framework that leverages bilinear pooling to capture cross-modal emotional inconsistency across audio-text and visual modalities. Specifically, the study seeks to answer the following questions:

1. To what extent does the proposed model's speech-text emotion recognition accuracy perform across the CREMA-D dataset.
2. To what extent does the proposed model's visual-only facial expression recognition accuracy perform across the CREMA-D dataset.
3. To what extent does the proposed multimodal deepfake detection framework using bilinear pooling fusion perform in terms of AUC on the FakeAVCeleb dataset?

Hypothesis

The hypothesis will serve as the anchor for the researchers' study. This study aims to test the following hypothesis to answer the main research questions:

Hypothesis 1: On Multimodal Deepfake Detection Performance

- **Null Hypothesis (H_0):** The proposed multimodal deepfake detection framework using bilinear pooling fusion does not achieve a competitive AUC score against existing detection frameworks on the FakeAVCeleb dataset.
- **Alternative Hypothesis (H_a):** The proposed multimodal deepfake detection framework using bilinear pooling fusion achieves a competitive AUC score against existing detection frameworks on the FakeAVCeleb dataset.

Scope and Limitation of the Study

Scope of the study

This study focuses on building and testing a deepfake detection system that detects deepfakes by checking the potential contradiction of information from two distinct sources: the audio-text channel, combining an individual's tone of voice and the content of what the individual is saying — and the individual's facial movements. Where other detection systems look for low-level visual artifacts such as blurry edges or abnormal skin tones, this system detects deepfakes by checking for inconsistencies between the two sources of information.

On the visual side, this study analyzes facial expressions through Action Unit saliency-guided keyframe selection, which prioritizes frames with high facial muscle activity to capture both sustained and brief facial expressions. The system checks whether the emotion shown on the speaker's face is consistent with the audio-text modality. The emotional scope of this study is limited to six universal emotion categories: anger, disgust, fear, happiness, sadness, and neutral. These six emotions were chosen as they are the most well-represented across the chosen training datasets and are widely recognized as the primary discrete emotion classes. Subjects with facial or vocal conditions that may affect the natural expression of emotion, such as facial palsy or speech disorders, are outside the scope of this study, as their

cross-modal patterns may differ significantly from the baseline emotional coherence the system is trained to recognize.

Delimitations of the study

In order to limit the scope of this study, several related topics are purposely not considered. First of all, this study does not aim to develop or improve any kind of deepfake generation technology. It is strictly detection-based, and no generation technology is developed or proposed here. The detection system also does not aim to detect deepfakes in real-time. It only operates on pre-recorded video files, not on live videos or live video chats. While videos uploaded to social media websites are within the scope, live detection of deepfakes is out of the scope of this research.

A critical input requirement of this system is that any video submitted for analysis must contain both a visual and an audio modality. Videos that lack either a clearly readable audio track or a detectable facial region cannot be processed by the system, as the detection framework relies on the interaction between both modalities to compute cross-modal emotional consistency. Videos that are audio-only, image-only, or in which the face is fully obstructed are therefore outside the scope of this research.

As far as what the system analyzes, it is purposely limited to two major modalities: the audio-text channel and the visual channel. The audio-text channel jointly analyzes the voice and spoken content, while the visual channel analyzes the face. Other possible modalities such as body language or eye movements are not considered. This is because facial expressions, speech, and language are the most obvious conveyors of emotional and semantic information, which is the basis of the detection method used in this research.

The detection system is also not meant to address scenarios in which a subject is purposely acting contradictorily to what they are saying, such as sarcasm or intentional hiding of

their emotions. Such scenarios may also result in a cross-modal inconsistency similar to what the system looks for in deepfakes, and it is not within the scope of this research to distinguish between the two. In addition, this research does not involve cyber tracking of the video's online origin to confirm where the video originated and how it was shared online, nor hardware forensic analysis to confirm the device that was used to record the original video.

Moreover, most of the data samples used in this study are from English speakers, since the chosen datasets — CREMA-D, MELD, and CMU-MOSEI — are mostly recorded in English. This means the model may not work as well on videos spoken in other languages.

Limitations of the research

Several limitations are acknowledged, which may affect the strength of the findings of this research. The most important limitation is that the model is dependent on the training data. The model will only be able to detect deepfakes that have some similarity to what the model has seen before, meaning the model may not perform well on deepfake styles or generation methods that are not included in the chosen datasets. This is especially important considering the rapid developments in diffusion models for generation purposes, as their ability to generate deepfakes with consistent emotions makes it a task that the model presented here cannot handle, since there are no deepfake emotion samples in any of the current benchmarks.

Another practical limitation is that the offline system is not applicable in scenarios that require real-time detection, such as live streaming or video calls, since the system is only applicable to pre-recorded videos.

The limitation of the system to two modalities also means that there are kinds of deepfakes that the system may not be able to detect. If a deepfake video is able to maintain a high degree of emotional consistency between its audio-text channel and visual channel, a goal that is becoming more and more achievable by sophisticated generative models, then the

system may not be able to identify it. Other behavioral modalities not analyzed in the scope of this study may be helpful in detection in such cases. Additionally, while the keyframe selection process is guided by Action Unit saliency to prioritize frames with high facial muscle activity, the system does not guarantee the detection of microexpressions, as these may occur between sampled frames.

Significance of the Study

This study aims to address the growing threat of sophisticated deepfake media by introducing a detection framework based on multimodal emotional inconsistency. The results of this study will be valuable to several key groups and institutions. Specifically, this research will significantly benefit the following:

Social Media Platforms. The viral nature of digital content means that manipulated media can reach millions of users within hours of upload, making early detection critical. The proposed framework could serve as a foundation for automated pre-distribution screening tools that help platforms identify and flag potentially manipulated audio-visual content before it spreads, reducing the impact of coordinated misinformation campaigns.

Artificial Intelligence (AI) Academic Community. This study contributes to the AI academic community by providing empirical evidence for the effectiveness of bilinear pooling in detecting cross-modal inconsistencies. Rather than simply asserting the superiority of intermediate fusion, the research demonstrates why modeling feature interactions outperforms standard concatenation in multimodal pattern recognition.

Law Enforcement and Digital Forensics Experts. As deepfakes are increasingly weaponized for fraud, blackmail, and identity theft, forensic investigators require detection tools that go beyond metadata analysis and visual artifact inspection. The behavioral and

cross-modal consistency analysis introduced in this study provides investigators with an additional layer of verification that is grounded in the biological properties of genuine human communication.

Journalists and News Organizations. Existing verification workflows rely heavily on visual inspection, reverse image search, and metadata checking. The findings of this study can help news organizations strengthen their content verification processes by highlighting cross-modal emotional consistency as an additional authenticity signal, thereby reducing the risk of inadvertently broadcasting manipulated media.

Definition of Terms

For the purpose of clarity and better understanding of the concepts used in this study, the following terms are defined both conceptually and operationally:

Deepfake

- **Conceptual:** Highly realistic synthetic video, audio, or images created using deep learning techniques to manipulate or replace a person's face or voice.
- **Operational:** The set of manipulated video files retrieved from the **FakeAVCeleb** datasets used as primary test subject, as well as the synthetic clips generated in this study through a four-track pipeline applied to the **CREMA-D** and **MELD** datasets.

Generative AI

- **Conceptual:** A branch of AI capable of generating new content, such as text, images, or video, based on patterns learned from existing data.

- **Operational:** The underlying technologies, including Generative Adversarial Networks, diffusion models, and neural voice conversion systems, responsible for creating the synthetic facial movements and vocal clones analyzed in this research.

Visual Artifacts

- **Conceptual:** Unnatural distortions or “glitches” in a digital image or video that results from deepfake generation.
- **Operational:** Low-level anomalies like flickering, blurring, or unnatural skin textures in a digitally manipulated video.

Multimodal Fusion

- **Conceptual:** The computational process of integrating information from multiple independent sources. Specifically the audio-text and visual modalities, to create a unified representation that enables more accurate decision-making than any single source could provide.
- **Operational:** The specific process of combining extracted features from the audio-text (voice and spoken content) and visual (face) streams into a single data vector for final classification.

Intermediate Fusion

- **Conceptual:** A fusion strategy where features are extracted from different sources separately and then combined at a middle stage of the neural network architecture.
- **Operational:** The specific layer in the proposed model where the audio-text feature vector Z_{at} (produced by concatenating Wav2Vec 2.0 and BERT outputs) and the visual

feature vector Z_v from the Vision Transformer are combined using Compact Bilinear Pooling to analyze emotional alignment.

Bilinear Pooling

- **Conceptual:** A sophisticated fusion operation that approximates the outer product of two feature vectors via Tensor Sketch (Fukui et al., 2016). Unlike simple concatenation, it captures all pairwise multiplicative interactions between features, allowing the model to detect complex, non-linear relationships across different modalities.
- **Operational:** The mathematical function implemented in the fusion layer of the model to calculate the relationship between audio-text and visual features, implemented as Compact Bilinear Pooling producing an 8,192-dimensional fusion vector, serving as the primary method for feature integration.

Cross-Modal Emotional Inconsistency

- **Conceptual:** A forensic marker in which the emotion conveyed in one source, like a happy facial expression, contradicts the emotion conveyed in another source, like an upset vocal tone.
- **Operational:** The numerical mismatch score computed as the absolute difference between the probability output of the audio-text emotion classification head and the visual emotion classification head within the detection framework, serving as an explicit indicator of cross-modal emotional divergence.

Multi-task Learning

- **Conceptual:** A training strategy in which a single model is simultaneously optimized for multiple related objectives, allowing shared representations to improve performance across all tasks.
- **Operational:** The training approach used in this study where the detection model is jointly optimized for binary fake/real classification and per-modality emotion classification, using a combined loss function across both objectives.

Emotion Head

- **Conceptual:** An auxiliary classification branch attached to an encoder that produces emotion category predictions from that encoder's feature output, used to impose emotion-discriminative structure on learned representations.
- **Operational:** Two lightweight fully connected layers attached to the audio-text representation and the visual encoder respectively, each producing a six-class emotion probability distribution used to compute the cross-modal inconsistency score and provide auxiliary supervision during training.

Discrepancy Score (Δ)

- **Conceptual:** A six-dimensional numerical measure of cross-modal emotional inconsistency, representing the per-class divergence between the audio-text and visual emotion probability distributions.
- **Operational:** The element-wise absolute difference between the softmax output of the audio-text emotion head and the softmax output of the visual emotion head, producing $\Delta \in \mathbb{R}^6$. This vector is concatenated with the 8,192-dimensional Compact Bilinear Pooling vector before the final classifier.

Precision

- **Conceptual:** The ability of the model to return only relevant instances by minimizing false positives.
- **Operational:** The percentage of videos flagged by the system as "Fake" that are actually deepfakes.

Recall

- **Conceptual:** The ability of the model to identify all actual deepfakes by minimizing false negatives or missed detections.
- **Operational:** The percentage of actual deepfakes in the test dataset that the model successfully identifies.

F1 Score

- **Conceptual:** The harmonic mean of precision and recall, providing a single metric that balances the trade-off between the model's ability to avoid false positives and its ability to capture all true positive deepfakes.
- **Operational:** A single numerical value ranging from 0 to 1 used to represent the overall effectiveness and reliability of the detection framework.

Area Under the ROC Curve (AUC)

- **Conceptual:** A threshold-independent metric measuring a classifier's ability to separate real from fake samples.
- **Operational:** The plotted area showing the system's ability to separate authentic media from deepfakes across various decision thresholds.

Chapter 2

REVIEW OF LITERATURE AND STUDIES

1. Limitation of Generative Deepfakes

Limitations in facial behavior

Although modern deep learning AI models can now generate a generally convincing adaptation of a human face, they often still struggle to replicate a human's facial behavior when it comes to expressing emotions or even just in communicating. Complex muscle movements create these microexpressions. However, usually, AI models are designed to produce media by making it more visually appealing to the audience instead of making it look real in terms of biological patterns of face movement. Verdoliva (2020) explained that usually, deepfake-generated media may look real but fail to mimic natural face movements such as those of the eyes and eyelids. As a result of such studies, facial irregularities can be used as a sign of manipulated media. Further studies made by Akhtar (2023) also indicated that AI systems have difficulties in maintaining a consistent facial motion when generating videos; this conclusion explains that deepfake generation models that are based on generative adversarial networks (GANs) may look visually appealing and realistic when viewed as static images but show irregular transitions when viewed in motion. These systems are trained to recreate facial appearances rather than the underlying biomechanics of a facial expression; the resulting videos may display unnatural facial movements or delayed reactions.

Another problem that the generative AIs have to contend with is the reproduction of genuine micro-expressions, which are involuntary facial movements that display genuine emotional responses. This is because the generative deepfake AIs have mostly been based on large datasets of facial images that may not have the fine details associated with the

expressions, leading to a slightly exaggerated, delayed, or inconsistent facial expression compared to a real human reaction.

Another study was done on the relationship between facial expressions and emotional consistency for its use in deepfake detection. Research on ACE-Net: A Fine-Grained Deepfake Detection Model with Multimodal Emotional Consistency (Yu et al., 2025) tells us that artificial intelligence may have a difficult time maintaining a consistent emotional image through facial expressions, further validating the claims made by the other studies reviewed above.

Limitations in lip-speech synchronization

One such important aspect of the natural communication of humans is lip speech synchronization, which is the synchronization of the lips' movement with the sound produced by the phonatory system. The coordinated movements of the vocal tract, including the lips, tongue, and vocal cords, are the phonemes. While it is now possible to create artificial facial movements and speech using deep learning techniques, achieving complete authenticity in speech patterns, which relies on lip-speech synchronization, remains an open challenge. This aspect can sometimes become evident in the case of deepfake videos, where even a slight inconsistency in the lips' movements with the sound produced by the lips may act as a criterion to judge the manipulation of the video. Deepfake technology mostly relies on the visual aspect of the speech pattern instead of paying more attention to the synchronization of the lips' movements with the sound produced by the lips (Tolosana et al., 2020).

Researchers have successfully used these inconsistencies as a key feature to detect deepfake videos. The studies by Dang et al. (2020) and Verdoliva (2020) indicate the capability of computer models to identify even minor discrepancies in the audio and lip movements, such as minor delays in the opening of the mouth, inappropriate shapes of the lips to produce specific phonetic sounds, and inappropriate changes in phonetic sounds. Tolosana et al. (2020) also

highlight the importance of the synchronization of lips with the speech, which, although imperceptible to the human eye, can be detected using multiple frames of videos. All the above studies suggest that the inability of deepfake models to synchronize audio-visual speech is an important indicator for detecting deepfake videos, indicating an inherent limitation of generative AI models.

Limitations in emotional consistency

Emotional consistency refers to the consistency between the facial expression, vocal tone, and text or speech conveyed in a message. This is a natural phenomenon in human communication. The emotions expressed by a human are always multimodal in nature. When a human is expressing happiness, the facial expressions, vocal tone, and text conveyed are always consistent with the expressed emotions. However, this is not the case with artificial intelligence systems used in deepfake media creation. The research conducted by Dang et al. (2020) on deepfake generators concluded that there are irregular emotional cues in the output generated by the system, leading to inconsistent output in the context. In a similar context, Hosler et al. (2021) concluded that the emotional inconsistency in the output of deepfake generators is due to the fact that emotional semantics are not effectively transferred from authentic videos to their deepfake counterparts, often causing inconsistency in the sentiment expressed in the audio and the facial expression.

Limitations in multimodal coherence

Multimodal coherence can be described as the natural congruence between the person's facial expressions and the tone of voice and language that they use while communicating. All these elements come into play in actual communication. Research by Hu et al. (2021) explains that videos that are manipulated frequently show inconsistencies between frames, and this occurrence is because generative AI models create approximate facial expressions. Similarly,

Tian et al. (2023) emphasize that deepfake videos commonly exhibit intra-modal and cross-modal inconsistencies between frames; from this, we can deduce that the information within a single modality (facial motion, for example) and across modalities (such as audio and visual signals) may not align naturally.

Zou et al. (2024) note that audio-visual deepfake detection systems often focus on the misalignment between the paired audio and visual signals, since synthetic media shows inconsistencies when speech and facial motion are generated separately or independently. In addition, Guo et al. (2025) tell us that deepfake detection models should be approached in a way that analyzes multi-domain inconsistencies because current generative AI still cannot reproduce relationships between modalities such as speech dynamics, emotional expression, and visual appearance.

Limitations in Temporal Consistency

Temporal consistency is the ability of a video to maintain stable natural changes across frames over time. Human behaviors such as head movements, blinking, and speech articulation all follow a continuous and predictable motion pattern. Deepfake generation does not present that in a natural way; the frames fluctuate, which is a good indicator of tampering. Deepfake systems fail to preserve realistic motion dynamics (Verdoliva, 2020). Tolosana et al. (2020) also emphasize that deepfake videos do not fully model the physical continuity of a human facial movement.

Several studies further underscore the fact that these temporal inconsistencies reflect the limitations of current technologies for generating deepfake media. For example, Dang et al. (2020) found that videos created through deepfake media generation often exhibit variations in the facial identity features or positions of the facial landmarks when the videos are analyzed frame by frame. Similarly, Gu et al. (2021) have shown that deepfake videos generated frame by

frame introduce spatiotemporal inconsistencies between adjacent frames and that these structural inconsistencies across frames can be exploited as reliable indicators of video manipulation. Moreover, Tolosana et al. (2020) have emphasized the fact that the deepfake media generation process often results in the generation of temporal artifacts, given the fact that these processes are more focused on generating realistic spatial information than on capturing the long-term motion dynamics of the objects being represented. As a consequence, it has been a fact that many of the recent techniques proposed for the detection of deepfake images concentrate on investigating the temporal information during a video sequence to identify the inconsistencies that are a consequence of the deepfake media generation process.

2. Datasets and Benchmarks for Multimodal Forensics

Through the partnership of Meta and Dolhansky et al. (2020), a large-scale forensic dataset named as DeepFake Detection Challenge (DFDC) served as one of the primary benchmarks for early generation artifacts. With 3,426 paid actors in a controlled environment, both real and deepfake-generated faces are augmented in a mega dataset, summing up to 128,154 ten-second clips (100,000 fake clips and 19,154 real clips). As this dataset focuses primarily on visual anomalies and generation artifacts, it still remains as the largest open-source face-swap dataset. To avoid bias by knowing how the manipulated data was produced, the researchers abstained from introducing their own deepfake detection model, thus sharing it to the community by making it into a competition with monetary benefit of the most accurate model. This dataset was structured into four distinct tiers:

- **Training and Validation sets:** Used to feed and fine-tune the participants' models.
- **Public Test set:** Used by the competitors to examine their performance against a public leaderboard during the contest.

- **Private/Hidden Test set:** Held and used by the organizers to formally evaluate model performance and determine the final rankings.

While the Training and Validation sets were released publicly for research, the Public and Private Test sets remained hidden to ensure the integrity of the Kaggle contest scores. However, while Dolhansky et al. argued that DFDC-trained models could generalize to real-world data due to its large volume, the dataset's "lab-generated" nature limits its versatility to adapt to the fast-changing state of state-of-the-art deepfake generative models. Another noticeable gap is its compositional nature, as their method of "faking" a sample is just stitching the voices and facial features from Actor A to Actor B, and vice versa. Thus, this artificial way of synthesizing deepfakes defeats the purpose of combating deepfake generation, as it only focuses on an observable box.

FakeAVCeleb Dataset

The FakeAVCeleb dataset, proposed by Khalid et al. (2022), is designed to overcome the deficiencies of previously used deepfake databases through the use of facial and voice manipulations. In contrast to existing benchmarks like FF++ and Celeb-DF, which focus on face-swap attacks, the FakeAVCeleb dataset includes three kinds of manipulated audio-visual pairs that have been synthesized via four different techniques, accompanied by fine-grain labeling specifying whether only the audio/video or both had been changed. The database is composed of 500 real audio-visual video sequences extracted from VoxCeleb2, along with 19,500 manipulated sequences corresponding to 500 celebrities of various races, overcoming the problem of racial bias observed in previous data sets. Importantly, FakeAVCeleb represents one of the first publicly available deepfake benchmarks to provide synthesized lip-sync fake audio together with their audio-visual deepfake counterparts. The multimodal composition of this database makes it useful for the purposes of developing an emotion-based deepfake detection

technique. Particularly, a special subset called voice cloning represents face manipulation through which the face is not altered while the audio is modified using the voice of another speaker, thus causing cross-modal inconsistencies that take the form of conflicting emotions of audio and video.

While less extensive than DFDC (19,500 vs. 128,000 videos), the FakeAVCeleb dataset possesses several notable strengths over its competitors: multiple types of manipulation, and a well-balanced distribution between visual and vocal modes. The FakeAVCeleb dataset is publicly accessible and has become a popular benchmark in recent research on multimodal deepfake detection, for example, Ilyas et al. (2024)

Although the FakeAVCeleb dataset serves as our basis for evaluating the performance of our proposed detection framework, it does not address the core training requirement of this study. The models to train need a solid foundation of how real humans express across different modalities, the model must learn various human expressions. To properly tackle the core requirements of this study, the researchers require datasets that focus primarily on human emotions and cater to both audio and visual modalities. Cao et al.'s (2014) Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D) research fills this gap by expanding the multimodal emotional scope with 7,442 clips from 91 diverse actors (48 male and 43 female). It provides unique and independent ratings for audio and visual channels, incorporating six common emotions: Anger, Disgust, Fear, Happy, Sad, and Neutral. This dataset was recorded in a controlled and observable environment with a solid green-screen background, with actors that are highly diverse according to ethnicity, age, and gender. The emotion labels and real-value intensity values for the perceived emotions were validated using crowdsourcing from 2,443 raters. Additionally, each actor was directed to utter 12 different sentences, meaning they used constant neutral dialogue acted out with the desired emotion. As a pioneer dataset for emotional multimodal analysis, it still remains relevant for recognizing preliminary human emotions.

However, this dataset alone cannot be used as a standalone training set for a model due to its outdated and lab-generated nature. Thus, it must be augmented with another dataset to fill the gap of its predictable environment. Furthermore, it is critical to include humans acting differently on purpose, such as being sarcastic, which is a huge gap in this dataset since it is confined to only six emotions and 12 constant sentences.

Another study by Poria et al. (2019), the Multimodal Emotion Lines Dataset (MELD), fixes this gap by incorporating 13,708 utterances from 1,433 dialogues from the TV series *Friends*, featuring seven different emotions such as Anger, Disgust, Fear, Joy, Neutral, Sadness, and Surprise. Along with these emotions, each video sample is sub-categorized into positive, negative, and neutral sentiments. As a challenge to the proposed training model, apart from the utterances, it includes interactions with other actors and background noises such as sitcom laughter tracks and music. This expands the scope of the emotional multimodal dataset compared to CREMA-D, which was orchestrated solely for controlled, independent acting. Having mixed distortions is useful to train models to detect human emotions; however, since MELD is completely based on a scripted television show, it still lacks the natural and unscripted authenticity of “in-the-wild” human expressions. If the model primarily focuses on the same quality, actors, and environment, the trained model will just be complacent in detecting if it is real or not.

Finally, to bridge the remaining gap of authenticity, a study by Bagher Zadeh et al. (2018) introduced the Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI) dataset. As the largest emotion-based multimodal dataset to date, it features over 23,000 annotated video segments from more than 1,000 distinct speakers sourced directly from YouTube. CMU-MOSEI’s closeness to real-world human conversations promotes authenticity which is guaranteed due to its “in-the-wild” nature unlike CREMA-D’s controlled environment or MELD’s scripted television acting. The dataset is rated and annotated for both

sentiment intensity and six universal emotions such as Anger, Disgust, Fear, Happiness, Sadness, and Surprise. Together, as we augment all CREMA-D, MELD, and CMU-MOSEI, we are able to generate a resilient dataset to train the models according to its purpose of detecting semantic and emotional dissonance in generated deepfakes.

3. Deepfake Generation Techniques

Deepfake generation models generally require many video and image datasets to increase the quality of its output (Abbas & Taeihagh, 2024); Hence, the typical target of deepfake videos are usually celebrities or famous personalities online. According to Kolagati et al. (2022), deepfake generation techniques use deep neural networks (DNN) to extract input features and generate hyper-realistic videos. Deep neural networks (DNN) is an architecture that is designed to mimic the structure and the functioning of the human brain, consisting of multiple layers that work simultaneously to learn patterns from data (Masood & Ahmad, 2023). Generative adversarial network (GAN), on the other hand, is a machine learning model that learns patterns from training datasets (Varughese, n.d.) and is composed of two competing DNNs (generator and discriminator). GANs are typically used to generate realistic data inferred from the patterns it has learned. Abbas & Taeihagh (2024) grouped deepfake generation into two methods: (1) Face reenactment & faceswap generation using generative adversarial network (GAN) variants and (2) Face and video synthetic generation using GAN variants.

1. Face reenactment & faceswap generation:

Face swapping refers to the method of swapping an individual's face from an input video to another individual's face from a different video (Ding et al., 2020). This process consists of three stages (Abbas & Taeihagh, 2024):

- a. The face from the input image will be detected and then another image from a library that is proportional to the original image will be chosen.

- b. After the original face is detected and the target image is decided, the facial features of the target image are swapped and adjusted to match the features of the source image.
- c. Finally, a combined candidate substitution is performed, which calculates distances based on the overlapping areas.

However, this method is prone to image appearance loss where some of the source image's features are lost. This makes the generated output look unnatural and unrealistic.

Face reenactment refers to the swapping of facial dynamics from a source image to a target image. This enables the target image to mimic the expressions of the source image. Unlike face swapping method, face reenactment focuses on manipulating facial attributes such as gaze, expression, and mouth movements (Tran et al., 2019; Bounareli et al., 2022). Gaze reenactment refers to the adjustment of the target's eye direction and positioning based on the source image. Mouth reenactment synchronizes the target's lip movement with the source input. Additionally, expression reenactment enables the transfer of emotional states, wherein the facial expressions of the source image drive those of the target.

2. Face and video synthetic generation:

This method refers to the process of generating faces/images that do not exist based on trained data (Abbas & Taeihagh, 2024). Generative adversarial networks have been successful in creating hyper-realistic facial images (Patel et al., 2022). In the context of video synthesis, the primary objective is to produce faces that do not exist in reality but appear indistinguishable from real people. The model DF-UD (facial and video synthetic generation-based model) achieved an accuracy rate of 91.08% on the FDC Facial Attribute dataset. However, DF-UD reported limited performance on low-level artifact features.

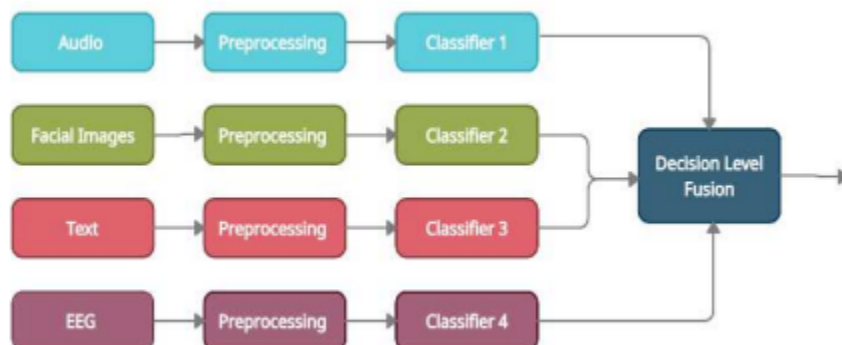
According to Kolagati et al. (2022), the two common techniques for deepfake generation are autoencoders and GAN. Examples of these are FakeApp, DeepFaceLab, DFaker, and DeepFake-tf.

4. Fusion Methods

Since combining different modalities for emotion recognition has proven to achieve higher accuracy, performance, and reliability than unimodal models (Agarwal & Desai, 2021), it is also equally important to choose the most appropriate fusion strategies as it could affect the performance of your model. Fusion technique refers to the process of combining information from different modalities, this may include data from text, images, audio, or/and video (Li et al., 2024). Fusion strategies aim to achieve a more comprehensive feature extraction by exploiting information across multiple modalities (Li et al, 2024). Karani and Desai (2022) classified multimodal fusion techniques into three: Decision-level, Feature-level, and Hybrid. For deep multimodal learning, an approach called intermediate fusion is commonly used (Xu et al., 2019).

1. Decision-level fusion: This approach processes each modality separately and combines the individual results through algebraic rules (Thakur et al., 2025). The fusion or the combination of results happens at a later stage (after classification of data), hence why in some references this fusion technique is referred to as ‘Late fusion’. Karani and Desai (2022) laid down the general framework of MER decision-level fusion technique for audio, visual, text, and EEG data.

Figure 2. Decision-Level Fusion Strategy Framework

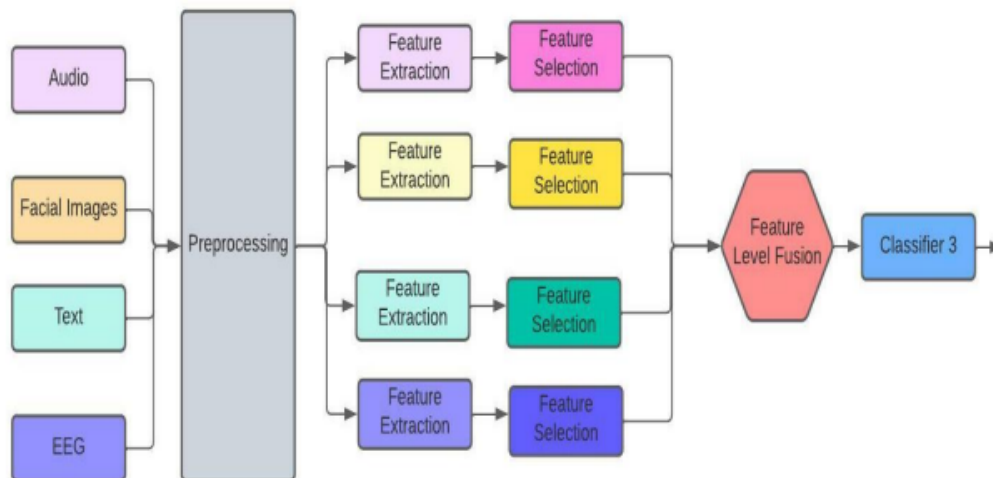


Note. From “Review on Multimodal Fusion Techniques for Human Emotion Recognition”, by Karani, R. & Dr. Desai, S. (2022), (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 13, No. 10. DOI:10.14569/IJACSA.2022.0131035

Since this fusion processes each modality independently, it is the most flexible technique among the three as it enables the application of the most suitable classifier for each modality. However, this strength could also make the model overlook cross-modal interactions (Wu et al., 2025). Various techniques that are employed on this fusion strategy include: Support Vector Regression (SVR), Blending algorithms, Brain Emotional Learning (BEL), Decision-tree based approach, Rule-based approach, Dempster-Shafer Theory (D-S), and Decision Template algorithm (DT).

2. Feature-Level Fusion: In this fusion strategy, features that are extracted from each modality are concatenated or combined using a different fusion technique before it is fed into the classifier (Karani & Desai). This captures feature-level interactions across each modality (Wu et al., 2025). A study conducted by Karani & Desai (2022), stated that although concatenation operation is generally used to implement feature-level fusion, it is not the best option for achieving good results. Furthermore, they also illustrated the general framework for this fusion technique.

Figure 3: Feature-Level Fusion Strategy Framework

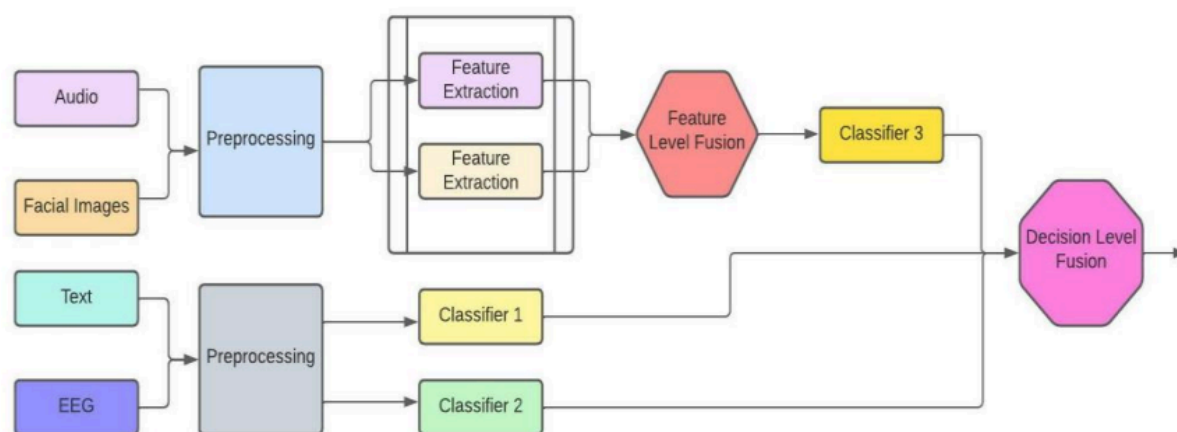


Note. From “Review on Multimodal Fusion Techniques for Human Emotion Recognition”, by Karani, R. & Dr. Desai, S. (2022), (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 13, No. 10. DOI:10.14569/IJACSA.2022.0131035

This fusion technique works by extracting the most discriminating features from each modality and eliminating repeated/irrelevant information (Karani & Desai, 2022) then combining them into a single feature vector (Thakur et al., 2025). Feature-level fusion is ideal for naturally connected and synchronized modalities like speech and facial expressions. This fusion technique faces challenges when modalities differ in temporal characteristics.

3. Hybrid Fusion: Hybrid fusion aims to combine the advantages of both the decision-level and feature-level fusion techniques. This method combines the result of feature-level fusion used on synchronous modalities and decision-level fusion on asynchronous modalities (Karani & Desai, 2022). Hybrid fusion is best suited for multimodal data inputs where some modality are synchronized in time and some modality are not. Karani & Desai (2022) also illustrated the general workflow of this fusion technique, as shown in figure 4.

Figure 4: Hybrid Fusion Strategy Framework



Note. From “Review on Multimodal Fusion Techniques for Human Emotion Recognition”, by Karani, R. & Dr. Desai, S. (2022), (IJACSA) International Journal of Advanced Computer Science and Applications, Vol. 13, No. 10. DOI:10.14569/IJACSA.2022.0131035

4. Intermediate Fusion: This approach balances the direct fusion of input data (feature-level) and the fusion of independent decisions (decision-level) (ApX Machine Learning, 2026). Intermediate fusion starts by processing each modality independently to extract relevant features. These features will be combined/fused into one shared representation (Xu et al., 2019). An article by ApX Machine Learning (2026) showed the different methods to perform this:

- **Concatenation:** as the name suggests, extracted feature vectors from different modalities are concatenated or joined end-to-end to form a single feature vector. If you have an audio feature vector v_{audio} and a text feature vector v_{text} , the concatenated result would look like $v_{fused} = concat(v_{audio}, v_{text})$. This unified feature vector contains information from all modalities.
- **Element-wise Operation:** This method can be achieved if the extracted feature vectors have the same dimension, the result will be the combination of all elements. Examples of these are summation, averaging, and multiplication (element-wise product).
- **More Complex Methods:** Advanced techniques like bilinear pooling allows multiplicative interactions between pairs of features across all modalities.

Once the modalities are combined into a single representation, it will then be processed by subsequent layers of a neural network.

Bilinear Pooling

Bilinear pooling has established itself as one of the most effective feature aggregation methods for fine-grained visual recognition tasks. Compared with common pooling methods, bilinear pooling can obtain better feature representation by capturing complex associations between high-order features. Its ability to model second-order feature interactions makes it particularly well-suited for tasks that demand discriminative representations, such as

fine-grained image classification, semantic segmentation, and medical image analysis (Zeng & He, 2022).

However, despite its representational power, standard bilinear pooling carries a critical practical limitation. Bilinear features are high dimensional, typically on the order of hundreds of thousands to a few million, which makes them impractical for subsequent analysis (Zeng & He, 2022). Specifically, the dimensions of bilinear representation are often up to hundreds of thousands or even millions. Its high dimensions entails significant computational and memory costs.

Compact Bilinear Pooling

These constraints have motivated the development of compact bilinear pooling (CBP), which seeks to retain the discriminative benefits of full bilinear pooling while dramatically reducing its computational footprint. Compact bilinear pooling methods address the problems of over-fitting and inefficiency that plague high-dimensional bilinear features (Arrieta, 2020). The key insight behind CBP is that the full outer-product computation can be approximated using randomized projection techniques.

In the multimodal classification domain, compact bilinear pooling is used for feature fusion to learn low-dimensional and expressive multimodal representations efficiently, addressing the heterogeneity and high-dimensionality issues that arise when fusing image and text features. This application, known as CBMLNet, was proposed by Li et al. (2024) and demonstrates the continued relevance of compact bilinear pooling in modern architectures.

The choice between full bilinear pooling and compact bilinear pooling ultimately hinges on the application's resource constraints and tolerance for approximation. Full bilinear pooling is preferred when computational resources are abundant and maximum representational fidelity is required. Compact bilinear pooling, on the other hand, is recommended when deployment

targets resource-limited environments, when features must be stored at scale, or when training time is a constraint (Sanchez et al., 2019).

5. Unimodal Detection Models

Video-Based Detection: XceptionNet and ResNext Architectures

Convolutional neural networks (CNN) have been at the forefront in the development of deepfake detection using video, with XceptionNet and ResNext having been cited as the most effective models. The XceptionNet model uses depthwise separable convolution for detecting fine-grained spatial features, which are a result of the face manipulation process. This makes the model more effective in detecting inconsistencies in face boundaries and texture patterns. The efficiency of the model in feature learning makes it suitable for detecting compression artifacts and blending features in deepfake videos. The ResNext model extends the success story of CNN models using its split-transform-merge model, which enables it to learn multiple feature representations at once (Gong et al., 2021).

Vision Transformers (ViT) for Facial Expression Recognition (FER)

Vision Transformers are recognized as a revolutionary technology for recognizing facial expressions, as they use self-attention mechanisms to incorporate both local facial features and global contextual relationships. Contrary to other CNN models that use hierarchical convolution to process the image, Vision Transformers process the image in patches to process the image in sequence, thereby including long-range dependencies that are necessary to process complex emotions. Results have shown that the latest implementation of Vision Transformers in FER has promising results, as they can achieve 100% accuracy in SAMM (Spontaneous Actions and Micro-Movements) and more than 95% in CK+ datasets (Jajan & Abdulazeez, 2024).

The application of ViT in the field of FER has shown some major advantages over the use of CNN-based models. A comparison of ViT, ResNet, and EfficientNet models in the field of FER showed that ViT-based models have an accuracy of 3-5% higher when compared to other models in difficult scenarios like the FER-2013 dataset. ViT-based models have shown high accuracy in scenarios involving different head poses and occlusions. This is due to the multi-head self-attention mechanism that can simultaneously focus on different areas of the face to learn what areas of the face are relevant to different emotions (Chen et al., 2022).

Audio-Based Detection: Wav2Vec and Transformer Models

The domain of audio detection has been revolutionized due to the introduction of self-supervised learning models, such as the use of Wav2Vec and Wav2Vec2. Wav2Vec2, being a transformer-based model, has been successful in learning robust speech representations using a large amount of unlabeled data, which carries linguistic as well as paralinguistic information required for authenticity detection. Wav2Vec2 has been successful in using the pre-training technique to learn robust features of speech at the acoustic level, which are generalized well across speakers of different languages and in various recording conditions (Hsu et al., 2021). The effectiveness of the Wav2Vec2 model has also been observed in detecting small-scale artifacts present in synthetic speech, such as those that are not detected using signal processing techniques, including unnatural prosody, emotional inconsistency, and timing problems (ParadaCabaleiro et al., 2023). Wav2Vec2 has been successful in demonstrating robustness to various types of noise and degradation in speech, along with maintaining high accuracy in the process of detection.

Wav2Vec2 for Speech Emotion Recognition (SER)

The Wav2Vec2 model has totally revolutionized the domain of speech emotion recognition by utilizing the self-supervised pre-training method, which maintains the linguistic

information and emotional prosody in the raw speech signal by utilizing the convolutional feature encoder and the transformer model. This is one of the major challenges in the domain of speech emotion recognition. From the empirical analysis carried out by Li & Zhang (2023), it is observed that the SER systems utilizing the Wav2Vec2 model have achieved state-of-the-art results in various benchmark datasets, thereby increasing the accuracy by 5-10% utilizing traditional feature-based methods.

The performance of the Wav2Vec2 model in emotion recognition is not limited to clean laboratory environments but can be applied to difficult real-world conditions. Research conducted to evaluate the performance of the Wav2Vec2 model in cross-corpus emotion recognition tasks indicates that the accuracy of the model, when fine-tuned, is very high, even when tested using a different dataset, as compared to CNN-based systems (Portal et al., 2025). In addition, the research conducted by ParadaCabaleiro et al., (2023), which was based on evaluating the performance of the emotion recognition system with nonsense speech, shows that the performance of the Wav2Vec2 model can be compared to that of humans, as it can detect emotions in the speech without taking into consideration the speech content.

The Limitations of Unimodal Detection Approaches

Although the current models are excellent in recognizing technical issues, the next breakthrough in detection will be emotional harmony. In other words, it is no longer enough for the lips to be in sync with the word. The emotion on the face has to be in harmony with the tone of the voice. This takes us away from analyzing digital issues and into the realm of analyzing human behavior. In this brave new world, an 'Emotion Mismatch' – i.e., someone saying something with an angry tone of voice, but their face shows no emotion – is the definitive proof that the video is fake.

The fast evolution of generative AI has introduced an unprecedented challenge in the

field of digital media authenticity. Until now, the conventional detection approaches have been grounded in unimodal analysis, in which either the visual or audio information has been taken into account in isolation. Nevertheless, as highlighted in the study by Cai et al. (2023), even these unimodal-based approaches are constrained by some inherent limitations, particularly in relation to the complex multi-aspect nature of human communication processes. As highlighted in the arguments by Heidari et al. (2023), another challenge that unimodal-based approaches have to face is associated with the issue of generalization in different datasets, and the fact that they are highly prone to sophisticated manipulation techniques. Nevertheless, the main limitation associated with unimodal-based detection approaches can be identified in relation to the fact that they fail to capitalize on the natural correlations that exist between different communication channels of natural content.

Due to the increasing complexities in deep learning-based deepfake generation, the research community has recently shifted towards multimodal detection techniques. The shift in the paradigm of deepfake detection techniques is based on the fact that, as human communication is real, it consists of synchronized modes. The multimodal detection techniques have been developed based on the fact that, although it is possible to generate various modes, it is extremely difficult to generate multiple modes at the same time (Javed et al., 2025). Based on the experimental results, studies have reported accuracy improvements ranging from 10 to 30% depending on the task and dataset.

Integration Frameworks: Combining Visual and Acoustic Features

One of the major challenges in the context of multimodal deepfake detection is the integration of features, which must be carried out in such a manner that it maximizes the feature complementarity as well as minimizes the computational cost. Recently, hierarchical fusion mechanisms have been implemented in deepfake detection systems, considering multiple levels of abstraction for feature fusion. The Novel Smart Deepfake Video Detection System, as

described by Elpeltagy et al. (2023), can be cited as an example where hierarchical fusion mechanisms are utilized for the detection of deepfake videos based on different levels of abstraction. In this context, the spatial features have been extracted using an upgraded version of XceptionNet for visual frames, while audio Constant-Q Transform (CQT) features have been extracted using a modified version of InceptionResNetV2. The features have been fused at the mid-layer level before passing through the Gated Recurrent Unit (GRU) with attention mechanisms for extracting temporal dependencies. The feature fusion mechanisms were found to have significant advantages in the context of deepfake video detection, where the video has been manipulated using only one modality, and the performance was found to be superior when compared to other state-of-the-art mechanisms using the FakeAVCeleb dataset.

6. Multimodal Emotion Recognition Models

Communication is fundamentally driven by emotion, including facial expressions, tone of voice, choice of words, and more. Emotions are complex and expressed through multiple nonverbal cues, which makes modelling human emotion a fundamental challenge in computing. Due to this complexity, it is difficult to build effective emotion recognition models based only on a single modality. Early studies on emotion recognition were mostly done based off of only a single modality, either visual or vocal features. Many have been successful in creating reliable unimodal models, while the *field* of multimodal models is still being widely researched. In multimodal emotion recognition systems, multiple data sources are combined to accurately model complex emotional states (Thakur et al., 2025), such as facial expressions, voice intonations, text information, body gestures, and more. Audio-visual recognition models have shown to improve the accuracy of speech recognition significantly. Especially in cases where background noise disrupts speech waveforms, lip movements help give visual cues to improve the recognition (Jeon & Kim, 2022). In a study conducted by Abdullah et al. (2021), they

concluded that MER models that employ deep learning generally achieved higher accuracy than unimodal models.

Most studies on MER models adopt a bimodal approach for data analysis. For visual feature extraction, Convolutional Neural Networks (CNNs) and Support Vector Machines (SVMs) are commonly employed, while Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) model are widely used for audio feature extraction (Ahmed et al, 2023). CNNs, LSTMs, and hybrid models like C3D-LSTM, and CNN-LSTM are one of the most widely adopted models in emotion recognition due to their ability to process complex emotional states (Thakur et al., 2022). In 2019, Noroozi et al. used a bimodal approach for data extraction and made an astounding 100% accuracy (under some limitations) by RF classifier using CNN, Multiclass SVM, and Random Forest on the Surrey Audio-Visual Expressed Emotion (SAVEE) dataset. Guanghui and Xiaoping (2021) achieved 98.92% accuracy on the eINTERFACE'05 BAUM-1 dataset using 2D-CNN for speech feature extraction, 3D-CNN for visual feature extraction, and SVM for classification. Wang et al. (2020) improved MER performance in comparison with unimodal systems using CNN, RNN, and LSTM on the RML, AFEW6.0, and eINTERFACE'05 datasets.

Table 1. Summary of bimodal emotion recognition methods.

Input Modality 1	Input Modality 2	Output Emotion	Fusion Strategy	Methods Used
Speech	Visual	Discrete emotions	Early fusion	CNN for visual, LSTM for speech
Speech	Visual	Continuous emotions	Late fusion	RNN for speech, SVM for visual
Visual	Physiological	Discrete emotions	Hybrid fusion	HOG for visual, CNN for signals
Speech	EEG	Arousal and valence	Early fusion	Spectrogram analysis, CNN
Text	Visual	Positive/negative	Attention mechanism	Transformer, attention network
Speech	Visual	Discrete emotions	Transfer learning	Pre-trained ResNet, RNN
Audio	Visual	Emotional state	Cross-modal Learning	CCA, CNN
Visual	Physiological	Stress detection	Hybrid fusion	LSTM for temporal, SVM
Speech	Visual	Empathy detection	Late fusion	CNN, random forest classifier
Text	Facial expressions	Sentiment analysis	Early fusion	TF-IDF for text, HOG for visual

Note. From “A Comprehensive Review of Multimodal Emotion Recognition: Techniques, Challenges, and Future Directions”, by Wu, Y., Mi, Q., & Gao, T., Biomimetics, 2025, 10(7), p.418 (<https://doi.org/10.3390/biomimetics10070418>)

Multimodal Emotion Recognition models typically follow the workflow illustrated in Figure 5 (Thakur et al, 2025). It begins by data acquisition (both audio and video data), followed by data preprocessing where the data goes through denoising and segmentation. After preprocessing, feature extraction will be done. Extracted features will then be combined using fusion techniques and fed into emotion recognition classifiers.

Figure 5. Typical MER Workflow



Note. From “A Comprehensive Review of Unimodal and Multimodal Emotion Detection: Datasets, Approaches, and Limitations”, by Thakur, P., Kaur, N., Aggarwal, N., & Singh, S. (2025), Expert Systems, 42(9). (<https://doi.org/10.1111/exsy.70103>)

Synthesis of the Study

Deepfake generation utilizes large datasets and deep neural networks, particularly GANs, to produce realistic synthetic media (Abbas & Taeihagh, 2024; Kolagati et al., 2022). However, based on the literature reviewed in this study, the capability of deepfake technology in mimicking human communication has been consistently shown to be wanting despite its technological sophistication. Verdoliva (2020) and Akhtar (2023) have conclusively demonstrated that the technology is incapable of mimicking the exact biomechanics of natural human facial movement. Yu et al. (2025) extended the same conclusion to the realm of emotional expression. Tolosana et al. (2020) and Dang et al. (2020) suggest that the mismatch

between lip movements and speech is one of the most reliable ways of detecting deepfakes, while Hosler et al. (2021) extended this to the realm of emotional semantics. Hu et al. (2021), Tian et al. (2023), Zou et al. (2024), and Guo et al. (2025) have cumulatively proved that the relationship between various modes of communication is independently created within deepfake technology, leading to the creation of inconsistencies, with Gu et al. (2021) and Tolosana et al. (2020) confirming the inability to uphold spatiotemporal consistency.

Until now, conventional detection approaches have been grounded in unimodal analysis. Convolutional neural networks (CNN) have been at the forefront in the development of deepfake detection using video, with XceptionNet and ResNext cited as the most effective models (Gong et al., 2021). The domain of audio detection has been revolutionized due to the introduction of self-supervised learning models, such as the use of Wav2Vec and Wav2Vec2 (Hsu et al., 2021; ParadaCabaleiro et al., 2023). Nevertheless, as highlighted in the study by Cai et al. (2023), these unimodal-based approaches are constrained by inherent limitations in relation to the complex multi-aspect nature of human communication processes. Heidari et al. (2023) notes they face challenges associated with the issue of generalization, and the main limitation is the fact that they fail to capitalize on the natural correlations that exist between different communication channels. Although current models are excellent in recognizing technical issues, the next breakthrough in detection will be emotional harmony; an 'Emotion Mismatch' is the definitive proof that the video is fake.

Due to these increasing complexities, the research community has recently shifted towards multimodal detection techniques (Javed et al., 2025). Multiple Emotion Recognition (MER) models combine multiple data sources from different modalities to accurately model complex emotional states, and according to Abdullah et al. (2021), MER models that integrate deep learning have higher accuracy than unimodal models. CNNs, LSTMs, and hybrid models are widely adopted in emotion recognition (Thakur et al., 2022), supplemented by technologies

like Vision Transformers (ViT) achieving highly promising results for facial expression recognition (Jajan & Abdulazeez, 2024; Chen et al., 2022) and Wav2Vec2 revolutionizing speech emotion recognition (Li & Zhang, 2023; Portal et al., 2025). To successfully implement this, fusion techniques are required. Karani and Desai (2022) classified multimodal fusion techniques into Decision-level, Feature-level, and Hybrid, while intermediate fusion combines extracted features into one single representation (Xu et al., 2019). When compared to other intermediate fusion approaches, CBP provides the best trade-off between representation capability and computation costs (Zeng & He, 2022). Hierarchical fusion mechanisms have shown significant advantages in deepfake video detection to maximize feature complementarity (Elpeltagy et al., 2023).

Ultimately, deepfake detection models must be properly trained using specific datasets. However, they lack the most important element in deepfake forensics: emotional inconsistencies. Therefore, this study concludes the chosen datasets will be an augmentation of CREMA-D (Cao et al., 2014), MELD (Poria et al., 2019), and CMU-MOSEI (Bagher Zadeh et al., 2018) to properly train the model with emotion dissonance tracking that can withstand various video qualities, conditions, and environments.

CHAPTER 3

METHODOLOGY

Research Design

This study follows a developmental-experimental research design, which is appropriate for studies that involve the construction and evaluation of the proposed system. The developmental component covers the design and implementation of the proposed multimodal deepfake detection framework, while the experimental component covers the systematic testing of the framework against existing frameworks using established dataset. Each stage of the study is structured to directly address a specific research question.

Research Question 1: Speech-Text Emotion Recognition Accuracy

Wav2Vec 2.0 and BERT are evaluated as auxiliary outputs of Emotion Head A after multi-task training across the CREMA-D and MELD datasets. Wav2Vec 2.0 processes the raw audio waveform to extract vocal emotional features, while BERT processes the transcribed text to extract linguistic sentiment. This isolates the speech-text modality to establish how well the proposed audio and text encoders perform together before they are combined with the visual modality.

Research Question 2: Visual-Only Facial Expression Recognition Accuracy

To address the second research question, the study develops and evaluates Vision Transformer (ViT) as auxiliary outputs of Emotion Head B after multi-task training on the same two datasets. This stage isolates the visual modality to establish a comparable performance baseline for the proposed visual encoder, allowing its independent contribution to the overall framework to be assessed before fusion is applied.

Research Question 3: Multimodal Deepfake Detection Performance

The complete framework is assembled by integrating the audio-text features (Wav2Vec 2.0 and BERT concatenated into Z_{at}) and visual features (ViT, Z_v) through Compact Bilinear Pooling fusion and evaluated on the FakeAVCeleb dataset using AUC as the primary performance metric. This stage tests whether the fusion of cross-modal emotional features produces a detection framework that is competitive with existing approaches, directly addressing the core computing problem this study aims to solve.

Sources of Data

This study relies on secondary data sources. The data used for this study was not collected firsthand by the researchers themselves, but rather gathered, compiled, and published by other individuals or organizations for their own research purposes and subsequently made available for use by the broader academic community. In the context of this study, these secondary data sources take the form of publicly available audio-visual datasets that have been established and validated by prior research in the fields of deepfake detection and multimodal emotion recognition. The primary data sources used in the researchers' project are the following:

- **FakeAVCeleb Dataset:** A multimodal deepfake database that consists of examples of face-swap as well as voice-cloning deepfakes having emotion tags. This dataset serves as the primary test set for evaluating the proposed framework's deepfake detection performance. In this study, FakeAVCeleb serves exclusively as the test set and is not used in any part of the training process.
- **Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D):** The authors have released this entire dataset on GitHub, making it fully open-source for academic

use, requiring approximately 7.5 gigabytes (GB) of storage space to clone the complete repository. In this study, CREMA-D serves as the fake generation source for Tracks 1, 2, and 3, with its emotion labels providing auxiliary supervision for the emotion classification heads during training.

- **Multimodal EmotionLines Dataset (MELD):** The benchmark was released publicly on Kaggle and is open-source and free to use for research and academic purposes, with a total size of approximately 11 gigabytes (GB). In this study, MELD is partitioned equally. One half provides real training samples, while the other half serves as the fake generation source for Track 4 through MuseTalk emotion-mismatch lip synchronization. Its emotion labels also supervise the emotion classification heads during training.
- **Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI):** The full set was released open-source on Kaggle with about 31 gigabytes (GB) of data across a wide variety of speakers. In this study, CMU-MOSEI provides real training samples exclusively for the binary fake/real classifier and is not used for emotion classification evaluation, as its labels are dimensional rather than discrete emotion categories.

Sampling Data

This study employs purposive sampling in the selection of datasets. Each dataset was deliberately chosen because it demonstrates specific characteristics that are directly relevant to the research objectives. The training datasets, CREMA-D, MELD, and CMU-MOSEI, were selected because they capture authentic cross-modal emotional expression across a wide range of speakers, recording conditions, and emotional categories, providing the model with a reliable ground truth for what genuine emotional coherence looks like across audio-text and visual modalities. Additionally, CREMA-D and MELD were selected as source material for data

augmentation, where a structured four-track deepfake generation pipeline is applied to produce synthetically manipulated training samples with known per-modality emotion labels. The FakeAVCeleb dataset was selected as the testing dataset for evaluating the proposed framework's deepfake detection performance, as it remains one of the largest and most widely used deepfake detection testers, allowing for direct and meaningful comparison against existing detection approaches.

- FakeAVCeleb:** As a result of having such extensive cover for multimodal manipulation detection, the presented dataset can be considered the main standard for assessing the effectiveness of the detection task within the framework under discussion. Consisting of 500 real videos with celebrities and around 19,500 manipulated videos, the FakeAVCeleb consists of four types of manipulations: face swapping, lip syncing, voice conversion, and a joint face-voice manipulation. Covering a diverse range of manipulation types, FakeAVCeleb serves as the primary benchmark for evaluating the proposed framework's deepfake detection performance. Thus, creating an authentic audio-visual mismatch of emotion, it becomes the main source of evidence for validation.
- Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D):** To establish a foundational understanding of authentic audio-visual alignment, Cao et al. (2014) introduced the Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D). This dataset fills the baseline gap by expanding the multimodal emotional scope with 7,442 clips from 91 diverse actors. Recorded in a controlled environment with a solid green-screen background, it provides unique and independent ratings for audio and visual channels across six common emotions.

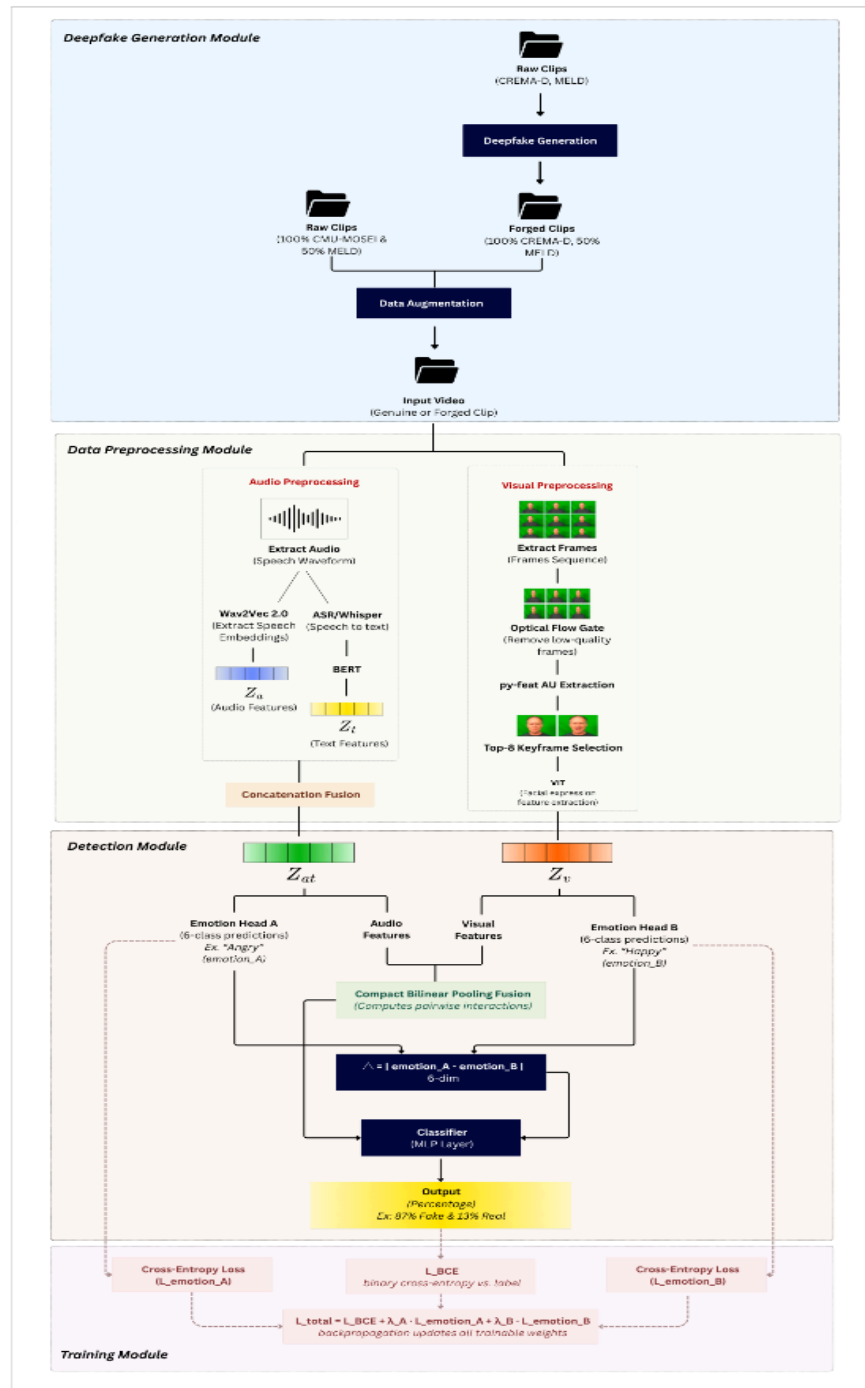
- **Multimodal EmotionLines Dataset (MELD):** While CREMA-D poses as a clean baseline, its high-fidelity controlled environment makes training models struggle to adapt to unpredictable changes in video quality. To address this discrepancy, Poria et al. (2019) proposed the Multimodal EmotionLines Dataset (MELD), featuring 13,708 utterances from 1,433 dialogues from the TV series *Friends*. It deliberately introduces challenges for the proposed model in the form of real-time acting interactions with other actors and background noise in the form of sitcom laugh tracks and music.
- **Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity (CMU-MOSEI):** To fill the remaining dataset gap of lacking of authenticity by MELD's scripted television acting, Bagher Zadeh et al. (2018) introduced the CMU-MOSEI dataset. One of the largest emotion-based multimodal datasets at the time of its publication, it includes over 23,000 annotated video segments from over 1,000 distinct speakers extracted directly from YouTube monologues, encouraging authenticity in the wild through its unscripted nature.

System Architecture

A. General Overview of the System Architecture

Figure 6. System architecture of the proposed emotional-based deepfake detection model

General Framework



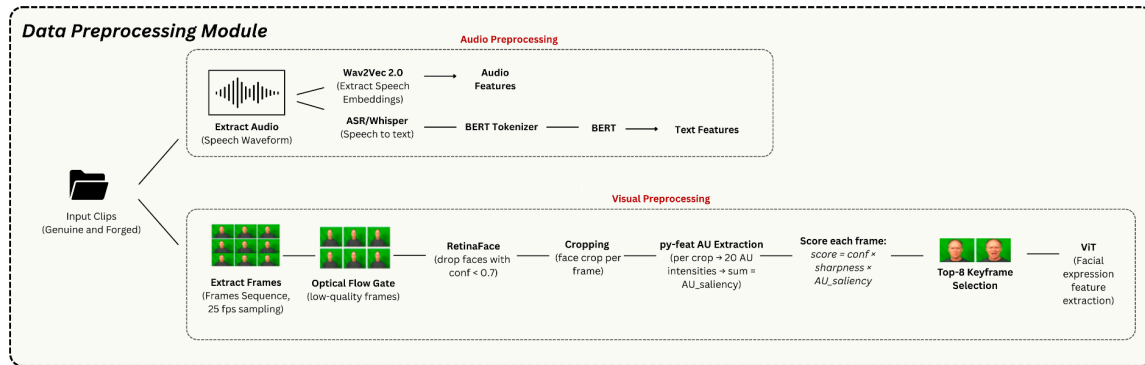
The system operates through four sequential modules: Deepfake Generation Module, Data Preprocessing Module, Detection Module, and Training Module.

- In **Deepfake Generation Module**, forged audio-visual pairs are synthetically generated from genuine emotional datasets to serve as training data.
- In the **Data Preprocessing Module**, raw video input is decomposed into its constituent modalities, audio waveform and visual frames, each of which is independently cleaned and prepared for feature extraction. Using three specialized encoders namely Wav2Vec 2.0 for audio, BERT for text, and Vision Transformer (ViT) for visual, the system independently extracts high-level emotional representations from their respective modalities. The audio and text features are then merged through concatenation to form a single audio-textual representation, while the visual features remain as their own representation, resulting in two grouped feature vectors that are passed on to the next module.
- After preprocessing, it will then shift to the **Detection Module**, where these two feature vectors are analyzed for cross-modal emotional consistency. Each of the two grouped vectors is fed into its own emotion classification head, producing six-class emotion predictions for both the audio-textual stream and the visual stream. In parallel, the two vectors are combined through bilinear fusion, which computes the pairwise interactions between them to capture how audio and visual features relate to one another. An emotion mismatch score is also calculated as the absolute difference between the two emotion predictions, providing an explicit numerical signal of cross-modal inconsistency. The fused bilinear representation and the mismatch score are then passed into a multi-layer perceptron (MLP) classifier, which produces the final binary prediction of real or fake along with a confidence score.
- Lastly, in the **Training Module**, the system is optimized using a combined loss function that balances three objectives: the binary classification loss for the real or fake prediction, and two emotion classification losses, one for each modality stream. These losses are summed together with adjustable weights and used to update the model's

parameters through backpropagation, allowing the system to simultaneously improve at detecting deepfakes and at recognizing emotion within each modality.

B. Data Preprocessing Diagram

Figure 7. Data Preprocessing Diagram



Given an input video clip, the preprocessing stage decomposes it into three parallel streams, audio, text, and visual, each processed independently before being passed to their respective encoders.

Audio Preprocessing Stream

The audio preprocessing stream begins with the extraction of the raw speech waveform from each video file. This waveform represents the raw acoustic signal of the speaker's voice and serves as the primary input to the audio processing pipeline. Once extracted, the pipeline splits into two parallel branches: the Wav2Vec 2.0 branch and the ASR-BERT branch.

1. Audio Preprocessing Branch

In the Wav2Vec 2.0 branch, the extracted speech waveform is resampled to a uniform rate of 16 kHz to match the input requirements of the Wav2Vec 2.0 encoder, which was pre-trained on 16 kHz audio. Unlike traditional audio models that require hand-crafted preprocessing such as log-Mel spectrogram conversion, Wav2Vec 2.0 is designed to process

raw audio waveforms directly through its internal convolutional feature encoder, which automatically learns robust acoustic representations from the raw signal without the need for manual feature engineering. This eliminates an entire preprocessing step and reduces the risk of information loss that can occur when converting a waveform into a fixed time-frequency representation. The resampled raw waveforms are fed directly into the pre-trained Wav2Vec 2.0 model, which extracts high-level speech representations encoding prosodic and emotional features such as pitch, energy, and speaking rate that are critical to the audio emotional analysis in this study.

2. Text Preprocessing Branch

In parallel with audio processing, the speech content of the clip is transcribed into text using an Automatic Speech Recognition (ASR) system. The resulting transcript is then tokenized using the BERT tokenizer, which internally normalizes casing to match the model's pretraining format. The tokenized text sequence is passed through BERT, a bidirectional transformer-based language model pretrained on large text corpora, which produces contextual word embeddings that capture the semantic and sentiment content of the spoken words. This textual stream complements the acoustic stream by providing a linguistic perspective on the speaker's emotional intent, what is being said, as opposed to how it sounds.

Visual Preprocessing Stream

All input videos are first standardized to a uniform sampling rate of 25 fps before any further processing, ensuring consistent temporal resolution across all clips regardless of their original recording conditions. Each input clip corresponds to a pre-segmented utterance. For MELD and CREMA-D, utterance boundaries are defined by the source datasets; for CMU-MOSEI, raw videos are pre-cut into utterance-level segments using timestamps from the official CMU-MOSEI annotation file and filtered to durations between 2 to 8 seconds. This

ensures that each clip captures a single, coherent emotional expression rather than spanning multiple emotional states, which would introduce ambiguity into both the emotion classification and the consistency detection tasks.

The resulting frame sequences are passed through a two-stage filtering pipeline that progressively narrows down the full frame set to only the most expression-relevant candidates.

In the **coarse filtering stage**, dense optical flow is computed between consecutive frames using the Farneback algorithm to estimate the magnitude of facial motion across time. Frames exhibiting mean optical flow magnitude below a threshold of 0.3 pixels are identified as static and discarded, since static faces carry minimal emotional information and would dilute the expressiveness of the feature representations. This stage efficiently eliminates large portions of redundant frames at low computational cost before any face detection is performed.

In the **fine-grained filtering stage**, the remaining candidate frames undergo face detection using RetinaFace, with detections below a confidence threshold of 0.7 discarded to ensure only clearly visible, high-confidence face regions are retained. Detected face regions are cropped from the bounding box provided by RetinaFace. Each candidate frame is then assigned an expressiveness score computed as the product of the RetinaFace detection confidence, Laplacian sharpness, and an Action Unit saliency weight derived from FACS-based facial muscle activity measurements obtained via py-feat. This AU saliency term prioritizes frames in which facial muscles are actively engaged, biasing keyframe selection toward microexpression-relevant moments rather than merely sharp or confident detections. The eight highest-scoring frames are retained as the final keyframe set for downstream processing. Clips in which face detection initially fails are processed through progressively relaxed detection thresholds to maximize the number of usable frames before downstream processing. All retained frames are resized to a standardized resolution of 224×224 pixels before being passed to the visual encoder.

C. Encoders

The system employs three specialized encoder models, one per modality, each used as a frozen feature extractor during Phase 1, with their pretrained weights held fixed while the emotion heads, fusion module, and classifier are trained on cached features, before being fine-tuned end-to-end during Phase 2.

1. Audio Encoder – Wav2Vec 2.0

Wav2Vec 2.0 serves as the audio encoder and is responsible for transforming the raw speech waveform into a high-dimensional emotional feature vector. It consists of a multi-layer convolutional feature encoder that maps raw audio into latent speech representations, followed by a transformer network that contextualizes these representations across the entire utterance. Because Wav2Vec 2.0 is pretrained through self-supervised learning on large unlabeled speech corpora, it has developed a deep understanding of speech structure — capturing not only phonetic content but also prosodic and paralinguistic cues such as pitch, energy, speaking rate, and emotional inflection that are directly relevant to emotion recognition. The resulting 768-dimensional acoustic embedding is concatenated with the BERT text embedding to form the joint audio-text representation (Z_{at}), described in the following section.

2. Text Encoder – BERT

BERT serves as the text encoder and processes the tokenized transcript produced by the ASR system. BERT's bidirectional attention mechanism allows it to model the full context of each word in the transcript, capturing sentiment, semantic meaning, and emotional language cues that complement the acoustic signal. The Wav2Vec2 and BERT outputs are concatenated into a single 1,536-dimensional vector (Z_{at}). EmotionHeadA takes this combined vector and predicts the audio-text emotion class.

There is no separate emotion head for the text encoder alone. The BERT output, combined with the acoustic features from Wav2Vec2, forms Z_{at} , a joint audio-text representation that captures both what was said and how it was delivered.

3. Visual Encoder – Vision Transformer (ViT)

The Vision Transformer serves as the visual encoder and processes the preprocessed facial image frames selected by the fine-grained filtering stage. Unlike CNN-based models that rely on local receptive fields, ViT divides each facial image into a sequence of fixed-size patches and processes them through a transformer with multi-head self-attention, enabling it to model both local facial features, such as the specific configuration of the eyes and mouth, and global contextual relationships across the entire face simultaneously. This makes ViT particularly well-suited for facial expression recognition, where subtle configurations of multiple facial muscle groups collectively convey emotional state. An Emotion Head B is attached to the ViT output during Stage 1, supervising visual feature learning across the same six emotion classes. The resulting visual feature vector encodes the emotional content of the speaker's facial expression and serves as the second input to the bilinear fusion module.

D. Hardware Specifications and computational constraints

All experiments were run on NVIDIA's RTX 4050 graphics card having 6GB of VRAM. End-to-end fine-tuning of all the backbones together was not possible for the computer. Backbones were kept frozen while training and features from the previous extraction phase were cached for use. The entire process took around 30 to 57 hours of computation.

E. Hyperparameter justification

- **Number of Keyframes (K=8):** In accordance with ACE-Net's coarse to fine keyframe extraction strategy. For 2-8 seconds videos at 25 fps, 8 keyframes suffice to mark the

start, peak, and end of an expression without repetition. Lesser keyframes could miss out on microexpressions while higher would only lead to unnecessary computational expense.

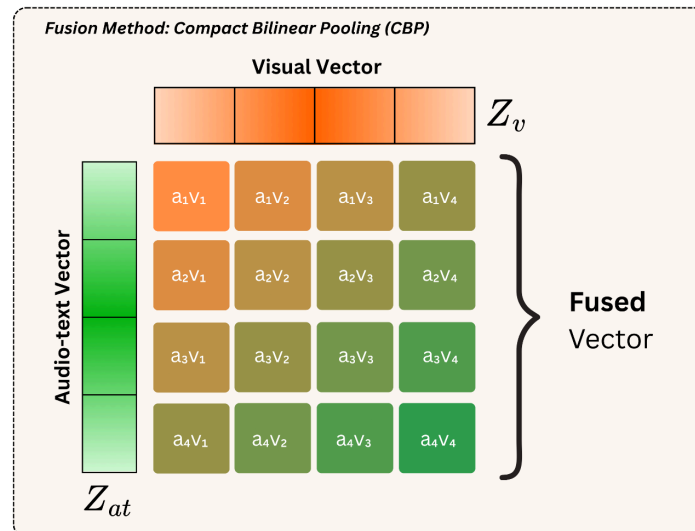
- **Weight Loss Constants ($\lambda_A = \lambda_B = 0.5$):** Symmetric loss function since both the modalities are assumed to be equally important for mismatch detection tasks.
- **Optical Flow threshold (0.3):** Value found experimentally for filtering out static frames while keeping all frames that involve motion.
- **Dropout ($p = 0.3$ for emotion heads, $p = 0.4$ for the classifier):** Standard setting due to limited size of data set.

F. Train-Validation-Test split

Each dataset was divided into a **80-10-10 (train-validation-test)** split at the level of the video clip such that there was no speaker overlap between train-validation-data sets. The training set was utilized for optimization, validation for early stopping and hyperparameter tuning, and testing for evaluation. Stratification was done based on label (real or fake) and source pipeline to ensure balanced representation across all generation tracks.

G. Bilinear Pooling Fusion Method

Figure 8. Bilinear Pooling Fusion Diagram



After the audio and visual encoders produce their respective feature vectors, these representations are combined through bilinear pooling, an intermediate fusion technique that captures all pairwise multiplicative interactions between the elements of the two feature vectors.

To make bilinear pooling computationally tractable, this study employs Compact Bilinear Pooling (CBP) via the Count Sketch algorithm (Fukui et al., 2016), which approximates the full outer product without explicitly computing it. Each input vector is first mapped to an 8,192-dimensional sketch vector $\psi(\cdot)$ through a randomized hash projection, then the two sketch vectors are combined via element-wise multiplication in the frequency domain using the Fast Fourier Transform (FFT):

$$f = IFFT(FFT(\psi(Z_{at})) \odot FFT(\psi(Z_v))) \in \mathbb{R}^{8192}$$

where:

- $f \in \mathbb{R}^{8192}$ is the compact bilinear pooling vector
- $\psi(\cdot)$ is the Count Sketch projection mapping each input to an 8,192-dimensional sketch vector
- $Z_{at} \in \mathbb{R}^{1536}$ is the joint audio-text feature vector
- $Z_v \in \mathbb{R}^{768}$ is the visual feature vector
- $FFT(\cdot)$ and $IFFT(\cdot)$ denote the Fast Fourier Transform and its inverse
- $\odot$ denotes element-wise multiplication in the frequency domain

This produces an 8,192-dimensional fused vector that approximates the pairwise multiplicative interactions of the full outer product at a fraction of the computational cost and with no additional learned parameters in the fusion layer.

This approach is fundamentally more expressive than simple concatenation or element-wise operations because it explicitly models every possible interaction between individual audio and visual features. In the context of cross-modal emotional consistency detection, this is critical: emotional mismatch between modalities manifests as complex, non-linear relationships between audio and visual features that simple concatenation, which merely stacks the two vectors without modeling their interactions, cannot adequately capture. For example, a high activation in an audio feature dimension corresponding to angry prosody combined with a high activation in a visual feature dimension corresponding to a happy expression produces a specific interaction term in the bilinear product that the MLP classifier can learn to associate with emotional inconsistency. This ability to detect subtle, feature-level cross-modal conflicts is what makes bilinear pooling particularly well-suited for the emotional mismatch detection task at the heart of this framework.

H. Operational Definition of Discrepancy Score

- Let $p_A = \text{softmax}(\text{EmotionHead}_A(Z_{at})) \in \mathbb{R}^6$ be the audio-text emotion probability distribution over the six emotion classes {neutral, happy, sad, angry, fear, disgust}.
- Let $p_B = \text{softmax}(\text{EmotionHead}_B(Z_v)) \in \mathbb{R}^6$ be the visual emotion probability distribution over the same classes.

The discrepancy score Δ is defined as:

$$\Delta = |p_A - p_B| \in \mathbb{R}^6$$

where the absolute difference is computed element-wise. Each dimension Δ_i represents the magnitude of mismatch for emotion class i . Δ is concatenated with the compact bilinear fusion vector (8,192 dimensions) to form the final 8,198-dimensional input to the MLP classifier.

I. Classifier

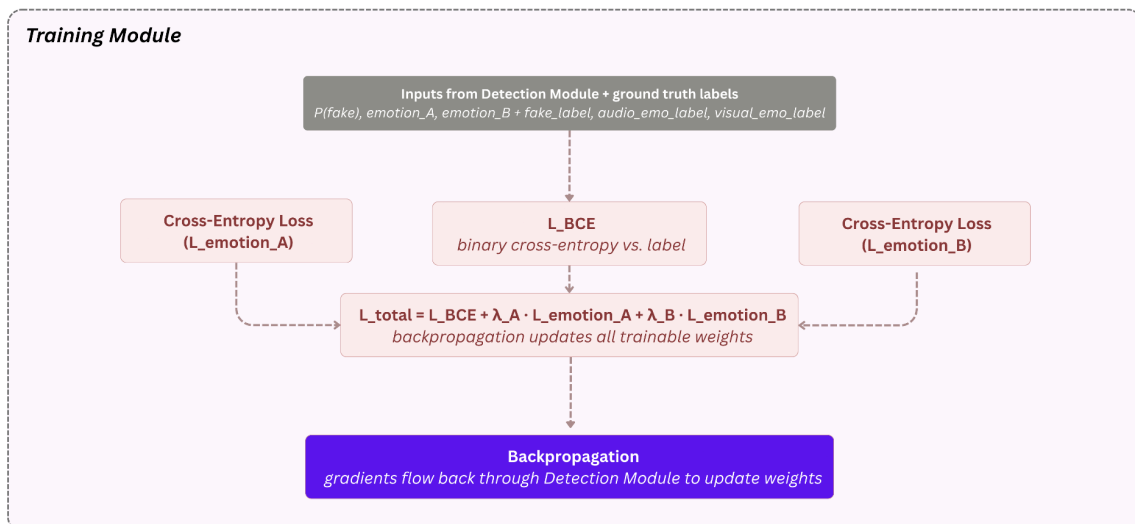
The fused bilinear pooling vector is passed through a multi-layer perceptron (MLP) classifier that learns a non-linear decision boundary separating genuine from forged audio-visual pairs based on their cross-modal emotional feature interactions.

The MLP consists of three fully connected layers with an inverted-triangle architecture: $8,198 \rightarrow 512 \rightarrow 128 \rightarrow 1$. ReLU activations and a dropout layer ($p=0.4$) after the first hidden layer prevent overfitting and improve generalization across diverse deepfake generation methods and recording conditions. The architecture progressively reduces the dimensionality of the fused representation, forcing the network to distill the most discriminative cross-modal consistency signals into an increasingly compact representation.

The final output layer produces a raw logit converted via sigmoid to a scalar probability score $p \in (0, 1)$ representing the model's confidence that the input clip is a deepfake. A score close to 1 indicates significant emotional inconsistency between audio and visual modalities, characteristic of forged content, while a score close to 0 indicates strong emotional coherence consistent with genuine human communication. At inference time, a decision threshold of 0.5 converts the probability score into a binary classification: real or fake.

J. Training

Figure 9. Training Module Diagram



The training module receives outputs from the detection module: $P(\text{fake})$, emotion_A , and emotion_B , alongside three ground truth labels: fake_label , $\text{audio_emotion_label}$, and $\text{visual_emotion_label}$.

Three losses are computed in parallel. L_{BCE} is binary cross-entropy between the predicted fake probability and the ground truth fake label. L_{emotion_A} and L_{emotion_B} are cross-entropy losses between the predicted audio and visual emotion distributions and their respective ground truth emotion labels. Emotion labels marked as unknown (-1) are masked and excluded from the loss computation.

These three losses are combined into a single multi-task objective:

$$L_{\text{total}} = L_{\text{BCE}} + \lambda_A \cdot L_{\text{emotion}_A} + \lambda_B \cdot L_{\text{emotion}_B}$$

where $\lambda_A = \lambda_B = 0.5$. This weighting ensures the model is jointly optimized for deepfake detection and cross-modal emotion consistency. Backpropagation flows through the total loss to update all trainable weights in the detection module.

Research Instrument

Given the computational and experimental nature of this study, the primary research instrument employed is a curated collection of open-sourced audio-visual datasets. Unlike traditional research instruments such as surveys or questionnaires, this study measures and evaluates its core variables, cross-modal emotional inconsistency, multimodal feature fusion via bilinear pooling, and deepfake detection accuracy, through structured audio-visual data that captures both authentic human emotional expression and synthetically manipulated media.

These datasets consist of audio-visual recordings of different actors delivering lines with different emotions and expressions. The set of emotions varies across datasets; however, common categories include anger, disgust, fear, happiness, neutrality, and sadness. These datasets will be divided into two, the first set will be used for training the model, while the second set will be used as a benchmark to test how well the model performs. The datasets that will be used for training include CREMA-D, MELD, and CMU-MOSEI. For testing, FakeAVCeleb will be used.

Contents of Datasets:

1. Training:

- **CREMA-D:** Consists of 7,442 audio-visual clips from 91 actors. In this dataset, there are 48 male and 43 female actors between the ages of 20 and 74. The actors' ethnicity is varied (African American, Asian, Caucasian, Hispanic, and some unspecified).
 - In CREMA-D, the actors spoke 12 sentences. Each sentence were acted out in different emotions (Anger, Disgust, Fear, Happy, Neutral and Sad) and in varying levels (Low, Medium, High and Unspecified)
 - CREMA-D serves exclusively as source material for deepfake generation in this study. CREMA-D clips serve as source material across three generation tracks, producing a combined 7,441 emotionally mismatched fake training samples through voice cloning, lip synchronization, and facial reenactment methods respectively.

- **MELD:** Consists of more than 13,000 utterances and 1,400 dialogues from the TV series 'Friends'. Each audio-visual clip is labeled with a specific emotion (Anger, Disgust, Fear, Joy, Neutral, Sadness, and Surprise).
 - MELD serves a dual role in this study: A subset of 3,334 clips are retained as authentic real-world training samples, while 3,482 clips serve as source material for cross-speaker lip-sync deepfake generation.
- **CMU-MOSEI:** Contains over 23,000 audio-visual clips from 1,000 speakers (comprising 57% male and 43% female speakers). In this dataset, the speakers were filmed while talking about diverse topics and monologues to capture how people naturally express emotions.
 - CMU-MOSEI is used exclusively as authentic real-world samples and does not contribute to deepfake generation, providing the model with diverse unconstrained real speech to anchor the real class.

2. Testing:

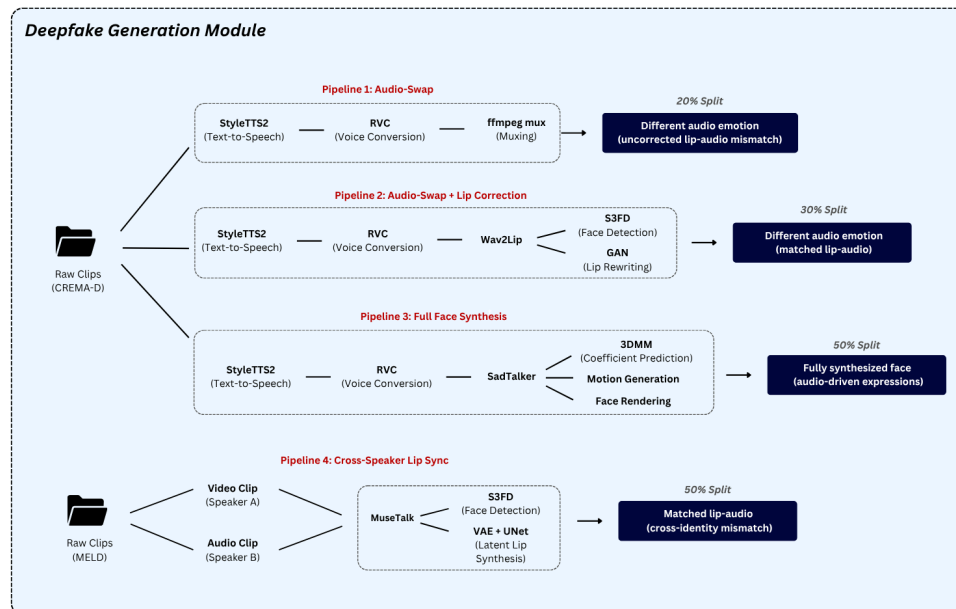
- **FakeAVCeleb:** Contains 20,000 audio-visual clips, comprising 500 real celebrity videos and 19,500 synthetically generated deepfakes derived from those videos. The real videos feature 500 celebrities across four ethnic backgrounds. Fake content was generated using four methods: FaceSwap, FSGAN, Wav2Lip, and SV2TTS, covering face-swapped and voice-cloned manipulations.

Data Generation/Gathering Procedure

All datasets used in this study are open-sourced and publicly available. The training datasets were obtained directly from their respective official sources and repositories. There are

other datasets that were extracted from Github as well. After data collection, the datasets are cleaned and standardized to normalize the data and account for missing values. This process will prepare the data for model training. Lastly, the original training datasets are fed into deepfake generation techniques to produce augmented data.

Figure 10. Data Augmentation Pipelines Diagram



To train the detection model to recognize emotional inconsistencies between audio and visual modalities, the study generates synthetic deepfake samples directly from the authentic training datasets, CREMA-D and MELD. Rather than relying solely on external deepfake

benchmarks for training, this internal augmentation strategy ensures that the model learns emotional mismatch as its primary and sole detection criterion, without being influenced by generation-specific artifacts from external tools.

The augmentation follows four distinct tracks, each producing a different type and severity of emotional mismatch:

Pipeline 1 – Audio Swap Only (CREMA-D 20%)

The simplest form of manipulation. The original audio track of a CREMA-D clip is replaced with a newly synthesized audio expressing a different emotion, generated using StyleTTS2 and converted to match the original speaker's voice using RVC. The face video remains completely untouched. Because the lips still move for the original emotion while the voice expresses a different one, this track produces the most obvious cross-modal mismatch and serves as the easiest case for the model to learn from.

Pipeline 2 – Audio Swap + Lip Correction (CREMA-D 30%)

An extension of Pipeline 1. After the mismatched audio is generated through StyleTTS2 and RVC, Wav2Lip rewrites the lip region of the original face video to visually match the new audio's phonemes. This corrects the most obvious artifact from Pipeline 1, the lip-audio mismatch, forcing the model to rely on subtler cues, specifically the disagreement between the upper facial expression (still reflecting the original emotion) and the audio emotion. This track represents a moderate level of manipulation difficulty.

Pipeline 3 – Full Face Synthesis (CREMA-D 50%)

The most sophisticated manipulation applied to CREMA-D. After generating the mismatched audio via StyleTTS2 and RVC, SadTalker takes a single portrait image from the original video and synthesizes a complete talking-head video driven entirely by the new audio. Every frame of the output is artificially generated, head pose, eye blinks, facial expressions, and lip movements are all produced by the model. Because SadTalker attempts to drive facial expressions from the audio signal, the emotional mismatch is less overt than in Pipelines 1 and 2, making this the hardest training case sourced from CREMA-D.

Pipeline 4 - Cross-Speaker Lip Sync (MELD 50%)

A fundamentally different type of manipulation that uses no speech synthesis at all. Two real MELD clips from different speakers are paired, the face video from one clip and the audio from another. MuseTalk then rewrites the lip region of the face video to match the paired audio. Because both the face and the audio are genuine human recordings, there are no TTS artifacts for the model to detect. The emotional mismatch arises purely from the fact that the two speakers were expressing different emotions, making this track the most realistic simulation of a real-world deepfake attack and the purest test of cross-modal emotional inconsistency detection.

Ethical Considerations

It should be emphasized that this study was developed from an understanding of the importance of both human rights and digital responsibility. The only type of data used in the project are those that are publicly accessible, and have been verified by third parties for their appropriateness in academic research. Data that is private, non-consensual, or otherwise sensitive will not be utilized in developing models which can detect such synthetic content. By

public data, the researchers are implying that consent was provided for the data to be utilized in a manner which supports the advancement of knowledge.

Preventing the misuse of synthetic media is the core approach to securing data. The manipulated video and audio used in testing the technology to detect synthetic content will be located in a secure environment, with access only to the researchers involved in this study. The technique is designed to enhance verification in the digital world rather than support deception in a more believable way.

Data Analysis

The data analysis procedure is structured according to the three research questions guiding this study. For Research Questions 1 and 2, the audio-text emotion head (EmotionHead A) and visual emotion head (EmotionHead B) are evaluated as auxiliary classifiers on the training data, measuring how accurately each head predicts the correct emotion class from its respective feature vector. Accuracy (ACC) is used as the primary performance metric, defined as the proportion of correctly classified clips out of all clips evaluated. This per-modality evaluation reflects the degree to which each feature representation captures discriminative emotional information prior to fusion.

For Research Question 3, the complete multimodal deepfake detection framework is assembled and evaluated on the FakeAVCeleb dataset. The outputs of the framework are analyzed using the Area Under the Curve (AUC) as the primary performance metric, measuring the framework's ability to distinguish genuine from forged audio-visual content across varying decision thresholds. AUC is chosen over accuracy for this stage because it provides a threshold-independent measure of detection performance, making it more suitable for evaluating and comparing deepfake detection frameworks across different operating conditions.

Statistical Treatment

The following statistical methods are employed to ensure that the results of the study are valid, reliable, and meaningful:

Accuracy (ACC). ACC is used as the primary metric for evaluating the performance of the speech-text and visual encoders on the emotion classification task in Research Questions 1 and 2. It is defined as:

$$\text{ACC} = (\text{TP} + \text{TN}) / N$$

where N is the total number of samples, TP is the number of correctly classified positive samples, and TN is the number of correctly classified negative samples.

Area Under the Curve (AUC). AUC is used as the primary metric for evaluating the deepfake detection performance of the complete framework in Research Question 3. It summarizes the ROC curve, which plots the true positive rate against the false positive rate across all decision thresholds, producing a single value between 0 and 1 that reflects the framework's overall ability to separate genuine from forged content regardless of the threshold applied.

Confusion Matrix. A confusion matrix will be built to provide a detailed summary of each condition's experimental identification results in four categories: true positives, which are deepfake identification results; true negatives, which are real identification results; false positives, which are real identification results; and false negatives, which are deepfake identification results. This metric will provide more granular results than can be obtained through any of the other metrics, such as accuracy alone.

Table 2. Confusion Matrix

True Positives A true positive occurs when the system accurately classifies the video as being a deepfake	False Positives A false positive occurs when a real video is wrongly identified as a deepfake by the model.
False Negative A false negative occurs when the system fails to detect a deepfake video and erroneously labels it as genuine.	True Negatives A true negative is an accurate detection of an authentic video by the algorithm.

Logistic Regression. A logistic regression classifier is implemented as a baseline for comparison against the proposed deep learning framework in Research Question 3. By using the same multimodal emotion features as input, this baseline isolates the contribution of bilinear pooling fusion over a simpler linear classification strategy.

Mean and Standard deviation

The mean and standard deviation represent two statistical measures that will be employed to characterize the results regarding the modality-level emotion consistency scores outputted by the system on the entire test set. The mean of each modality score for the audio-text stream (EmotionHead A) and visual stream (EmotionHead B), will illustrate a reference point, while the standard deviation scores indicate the range of each modality's scores for each medium. A high standard deviation on the deepfake class, for example, might indicate that AI-generated videos tend to have inconsistent emotional signals. It will also be used to describe some characteristics about the dataset, such as real and deepfake video counts on **FakeAVCeleb**.

Precision

The precision score is defined as the proportion of positive predictions that are actually correct.

$$Precision = TP / (TP + FP)$$

where TP is the number of True Positives and FP is the number of False Positives.

In this study's context, a high precision score means that the system seldom sends false alarms, i.e., the system rarely flags real videos as deepfakes. Precision will be reported at the single-modality level as well as at the fusion output. This will enable the study to analyze whether the proposed fusion decreases the false positive rate of the system.

Recall

The recall, which is also referred to as sensitivity, is a measure of how well the system performs in recognizing actual deepfake videos.

$$Recall = TP / (TP + FN)$$

where *TP* is the number of True Positives and *FN* is the number of False Negatives.

Here, a high recall value signifies that the system does not miss many fake videos. In that case, however, it may be flagging many real clips as well, which would result in a low false positive rate and hence, low precision. In natural deepfake detection pipelines with human raters in the loop, precision is typically more valued than recall, since it is favorable for a system to send few false alarms to reviewers, even if it means missing a few deepfakes.

However, recall is a significant metric for this study due to the potential repercussions in the real world of a false negative prediction for a deepfake video.

F1-Score

The F1-score is reported as a supplementary metric alongside AUC to provide insight into the precision-recall tradeoff of the framework. It is more appropriate when the costs of false positives and false negatives are high. When detecting deepfakes, the costs of false positives and false negatives are high. It is a great way to evaluate the system because it settles the trade-off between recall and precision values by averaging them as one value. The F1-scores will be reported for the audio-text modality (EmotionHead A) and the visual modality (EmotionHead B), as well as the full fused model output, to demonstrate that this system can be improved with the addition of bilinear pooling fusion.

ROC Curve and Area Under the Curve (AUC)

The receiver operating characteristic (ROC) curve is a plot of true positive rates and false positive rates at various thresholds, and AUC is a numerical value between 0-1 that summarizes the ROC curve. AUC near 1.0 indicates a highly performing model that can discriminate between real and deepfake videos regardless of the applied thresholds.

Bootstrap Confidence Interval (Bootstrap CI)

Bootstrap CI is a resampling-based statistical method that estimates the reliability and stability of a performance metric by repeatedly resampling the dataset with replacement and computing the metric on each resample (Efron & Tibshirani, 1993). It is used in this study because the proposed framework's AUC on the FakeAVCeleb dataset cannot be directly compared against existing frameworks through formal significance testing, as per-sample prediction scores from other models are unavailable. Bootstrap CI provides a statistically grounded alternative by estimating how stable and reliable the proposed framework's AUC is across different subsets of the dataset. It is applied by resampling the FakeAVCeleb test set with replacement across numerous iterations, computing the AUC on each resample, and constructing a 95% confidence interval from the resulting distribution. The confidence interval is then compared descriptively against the reported AUC values of existing detection frameworks to assess whether the proposed framework performs competitively.

Table 3. Summary of Statistical Tools and Their Application

Metric / Tool	Purpose in This Study	Where Applied
Mean & Standard Deviation	Summarize modality score distributions across datasets	Per-modality score analysis

Accuracy	Overall correct classification rate	Final fused model evaluation
Precision	Minimize false deepfake flags	Per-modality and fused model
Recall	Maximize deepfake detection rate	Per-modality and fused model
F1-Score (Supplementary)	Balance precision and recall; main performance metric	Per-modality and fused model
Logistic Regression	Baseline classifier for benchmarking	Baseline comparison
ROC Curve / AUC	Threshold-independent performance across datasets	Multi-dataset evaluation
Confusion Matrix	Identify specific error types and failure modes	Per-dataset diagnostic analysis
Bootstrap CI	Threshold-independent performance across datasets	Multi-dataset evaluation

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MOCK-UP

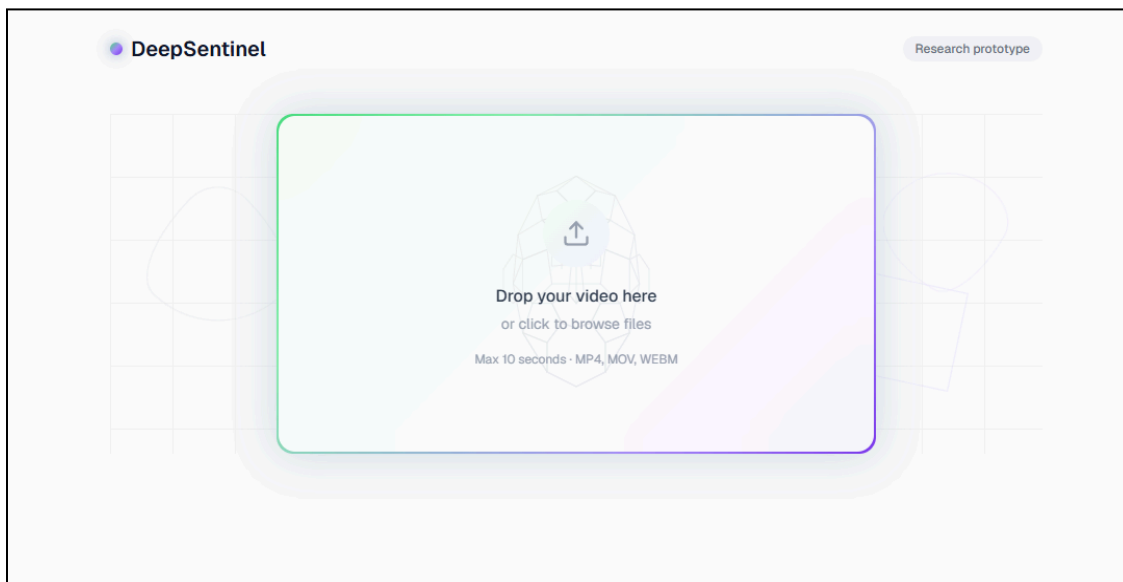


Figure 11: Front page that prompts user to upload a video

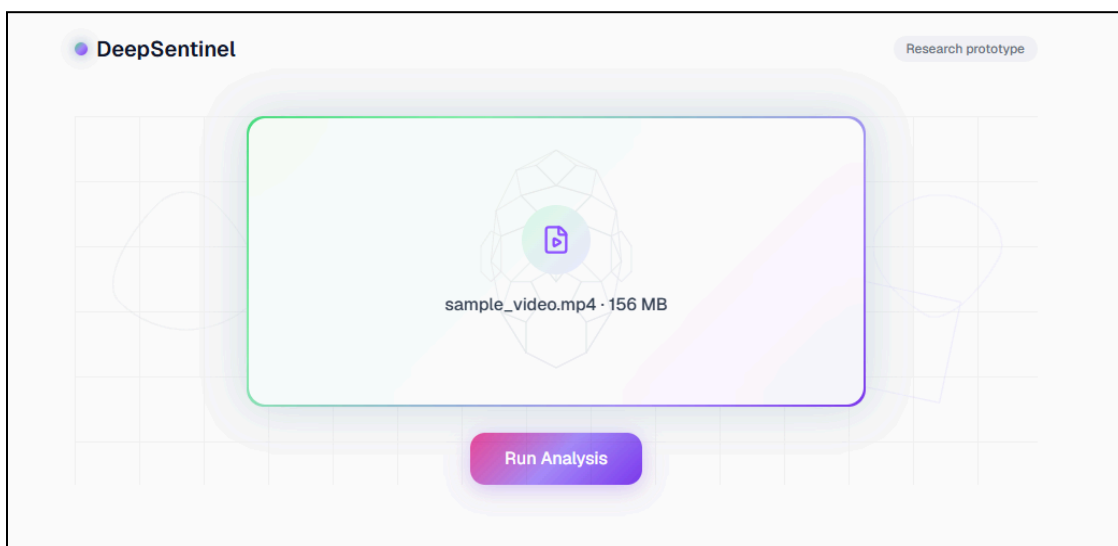


Figure 12: Interface when user uploaded the input video

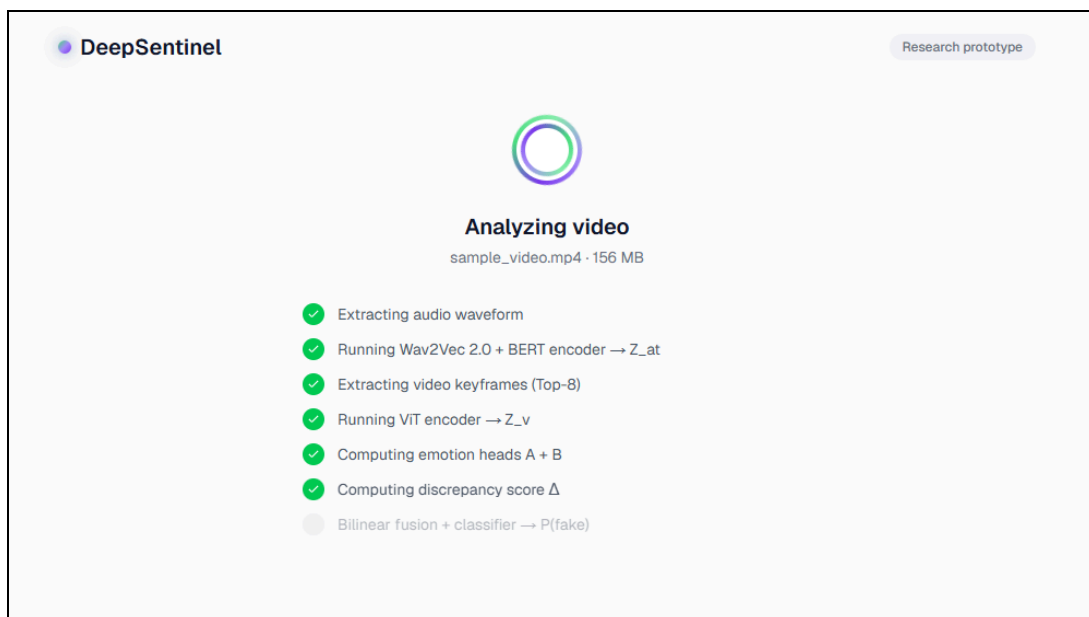


Figure 13: Analyzing process loading screen

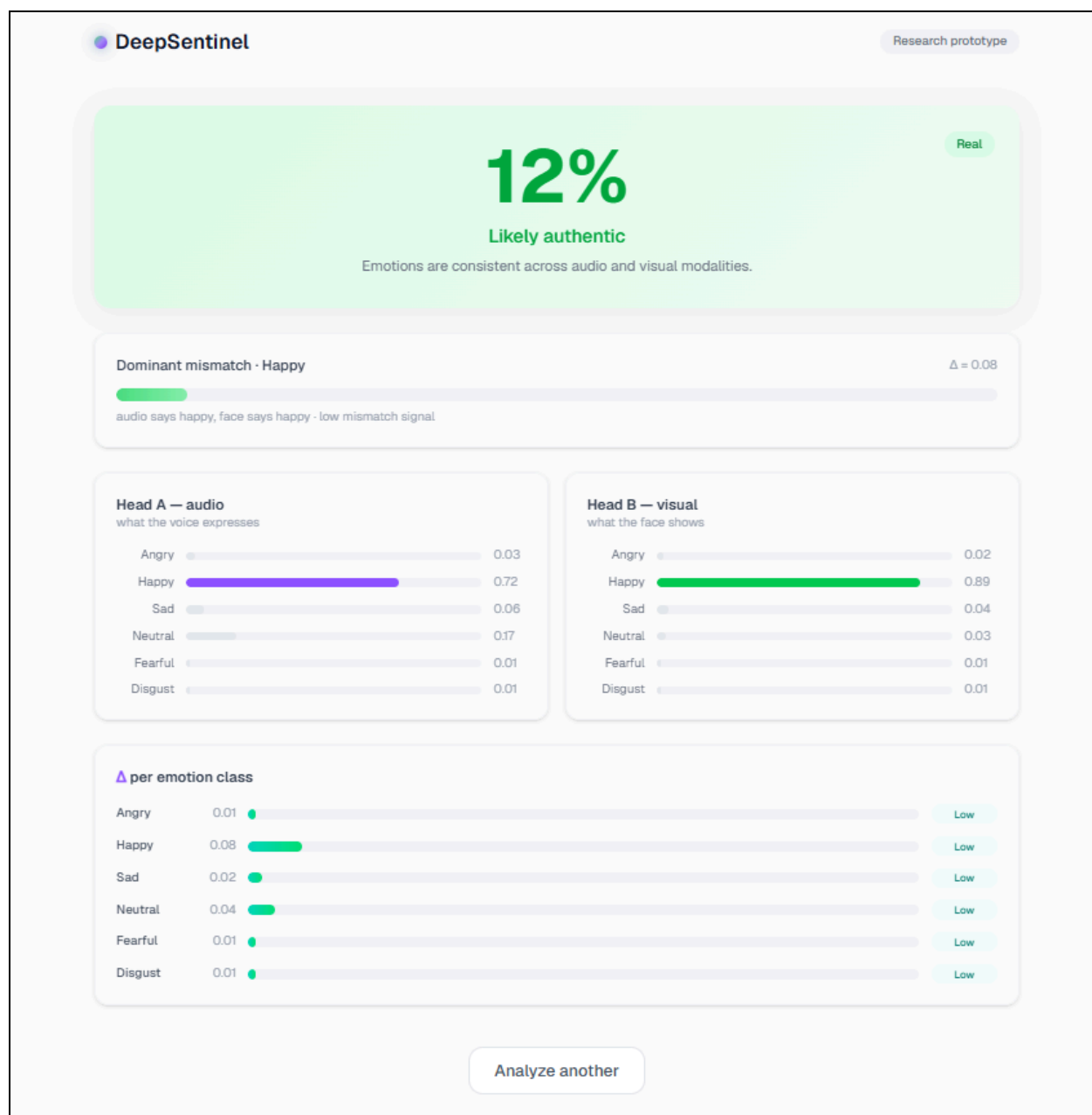


Figure 14: Successful analysis - when the video is authentic/real

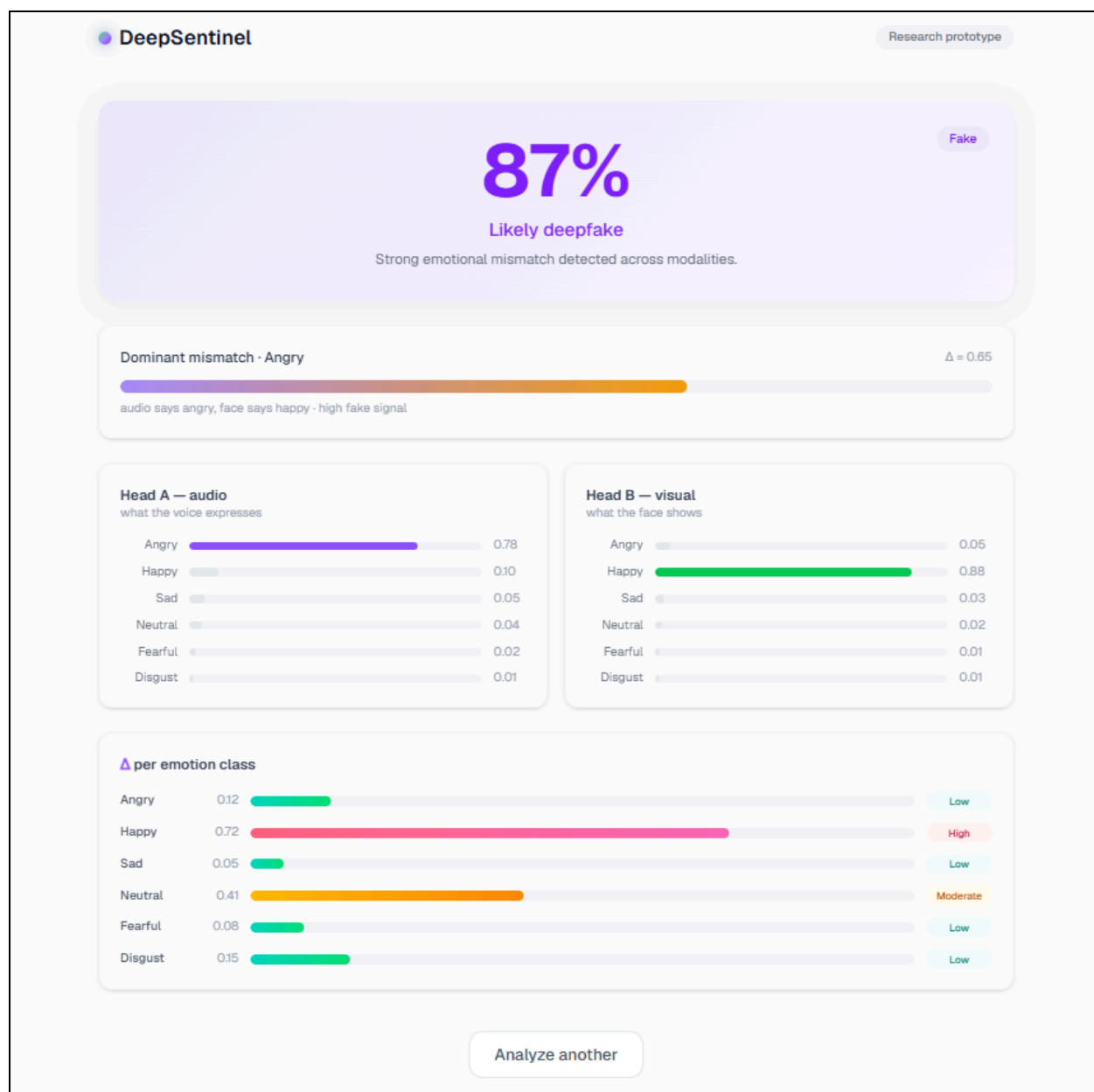


Figure 15: Successful analysis - when the video is fake